Distribution Theory

Functions, Measures, and Generalized Functions

A semi-formal introduction, with applications

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Preface

What these notes are

The theory of distributions is the part of analysis that lets us do the things calculus tells us we may not do: differentiate $|x|$, $\operatorname{sgn} x$ or an indicator function $\mathbf{1}_E$; call a discontinuous function a solution of a differential equation; take the Fourier transform of 1, of x, or of $\arctan x$; compute with the "Dirac delta function"; and even make sense of fractional derivatives. Instead of describing an object by its values at points, we describe it by how it *acts* on a family of very well-behaved *test functions*.

We develop this viewpoint in three layers. Ordinary (locally integrable) functions form the first layer. Radon measures form the second, and they already contain objects, such as the Dirac mass, that no function can represent. Distributions form the third and most general layer. Each layer is introduced as a generalization of the one before, and at each stage we ask the reverse question: when is an object of a higher layer secretly an object of a lower one?

The emphasis is deliberately on intuition and on examples. The topologies on spaces of test functions are described, but whenever possible they are replaced by concrete inequalities that can be checked by hand. A large number of examples are worked out in full, including several "obvious" ones. The second half of the book collects applications: ordinary differential equations, a two-line proof of Liouville's theorem, fundamental solutions for the transport, heat and wave equations, elliptic regularity, the Poisson summation formula, probability, the Weierstrass approximation theorem, and the distributional definition of perimeter.

Prerequisites

The first chapters require only a solid command of calculus and some comfort with integration. Later chapters assume real analysis (Lebesgue integration and basic measure theory), a little complex analysis, and some familiarity with partial differential equations. Readers at different levels will therefore take different things away; it is perfectly reasonable to read the first few chapters closely and skim the rest.

Acknowledgments

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Further reading

The standard references are Hörmander [1], the gentler Duistermaat–Kolk [2], and Part II of Rudin's *Functional Analysis* [3]. For perimeter and geometric measure theory see Maggi [4].

Conventions and Notation

None of the following needs to be absorbed before starting; this chapter is meant to be consulted as needed.

Notation

- j, k, m, n always denote non-negative integers, and N always denotes the dimension of the ambient Euclidean space $\mathbb{R}^N$.
- α, β denote ***multi-indices***: tuples $\alpha = (\alpha_1, \dots, \alpha_N)$ of non-negative integers. The ***order*** of α is $|\alpha| = \alpha_1 + \cdots + \alpha_N$. We write $x^\alpha = x_1^{\alpha_1} \cdots x_N^{\alpha_N}$ and

$$\partial^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \cdots \partial x_N^{\alpha_N}}.$$

  For instance, if $N = 3$ and $\alpha = (2, 0, 3)$ then $|\alpha| = 5$ and $\partial^\alpha f = \partial_x^2 \partial_z^3 f$.
- In any dimension, $\int_E f \, dx$ is shorthand for $\int_E f(x) \, dx$, and $\int f\phi$ is shorthand for $\int_{\mathbb{R}^N} f(x)\phi(x) \, dx$.
- $\operatorname{sgn} x$ equals 1 if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$.
- $\mathbf{1}_E$ is the indicator (characteristic) function of the set E.
- L^p means $L^p(\mathbb{R}^N)$, and similarly for $L^1_{\text{loc}} = L^1_{\text{loc}}(\mathbb{R}^N)$.
- The ***Fourier transform*** is normalized as

$$\hat{f}(\xi) = \mathcal{F}f(\xi) := \int_{\mathbb{R}^N} f(x) \, e^{-2\pi i x \cdot \xi} \, dx,$$
$$\check{f}(x) = \mathcal{F}^{-1} f(x) := \int_{\mathbb{R}^N} f(\xi) \, e^{2\pi i x \cdot \xi} \, d\xi.$$

  When an expression is too wide for a hat we write $(\cdots)^\wedge$.
- "Measure" almost always means (possibly signed) *Radon* measure.
- For a distribution T and a test function ϕ, $\langle T, \phi \rangle$ is another way of writing $T(\phi)$; see Section 2.1.
- D or L usually denotes a linear partial differential operator. Its coefficients will always be *constant*; much of the theory breaks down otherwise.
- $C_c^\infty(U)$ denotes smooth, compactly supported functions on U, and $\mathcal{S}(\mathbb{R}^N)$ the Schwartz functions (Section 1.3.2).

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I

Foundations

1 Three Layers of Generalized Functions

Study an object by studying how it acts on other objects.

- The guiding philosophy of *distribution theory*

We will build distributions as the top floor of a three-storey building. The ground floor consists of functions, the middle floor of measures, and the top floor of distributions. Each floor contains the one beneath it, and each contains genuinely new objects.

1.1 The first layer: functions

The word "function" in these notes almost always refers to an element of L^1 or L^1_{loc} on some domain. Strictly speaking such an element is not a function but an *equivalence class* of functions: two functions are identified when they differ only on a set of Lebesgue measure zero. This is harmless (and in fact natural), because integrals cannot detect changes on null sets, and it gives L^1 a much better structure.

Consequently it is meaningless, in general, to ask for the value of an L^1 function at a single point. Nevertheless every $f \in L^1_{\text{loc}}$ has a ***canonical representative***, obtained by averaging over shrinking balls:

$$\tilde{f}(x) := \lim_{r \to 0^+} \frac{1}{|B(x,r)|} \int_{B(x,r)} f. \tag{1.1}$$

By the ***Lebesgue differentiation theorem*** this limit exists and equals $f(x)$ for almost every x. This lets us speak about pointwise properties of L^1 functions: we say f is non-negative, or continuous, if its canonical representative is.

1.2 The second layer: measures

Measures formalize the idea of assigning a size to regions: area, volume, total mass, or total electric charge (which can be negative). Because we work only in $\mathbb{R}^N$, we need only one kind of measure.

Radon measures

The following notions will be used throughout.

- The ***Borel σ-algebra*** $\mathcal{B}(\mathbb{R}^N)$ is the smallest family of subsets of $\mathbb{R}^N$ containing the open sets and closed under complements and countable unions. Its members are ***Borel sets***.
- A ***Borel measure*** is a map $\mu : \mathcal{B}(\mathbb{R}^N) \to [0, \infty]$ with $\mu(\varnothing) = 0$ which is *countably additive*: $\mu\big(\bigcup_n A_n\big) = \sum_n \mu(A_n)$ for pairwise disjoint Borel sets $A_1, A_2, \ldots$. It is ***finite*** if it never takes the value ∞.
- A ***signed Borel measure*** is a difference of two Borel measures, at least one of which is finite.
- A (signed) ***Radon measure*** is a (signed) Borel measure that is finite on every compact set.

Remark 1.2.1. Some of these are, strictly speaking, characterizations rather than the textbook definitions, but they are correct descriptions of the objects in question. From now on "measure" means "Radon measure, possibly signed".

1.2.1 Every function is a measure

A function $f \in L^1_{\mathrm{loc}}(\mathbb{R}^N)$ determines a signed Radon measure

$$\mu_f(E) := \int_E f \, \mathrm{d}x.$$

Distinct functions give distinct measures, so the assignment $f \mapsto \mu_f$ is injective, and we may regard measures as generalized functions.

It helps to picture f as a mass density. In the classical picture, f assigns a *value* to each point. In the new picture, f assigns a *mass* to each region.

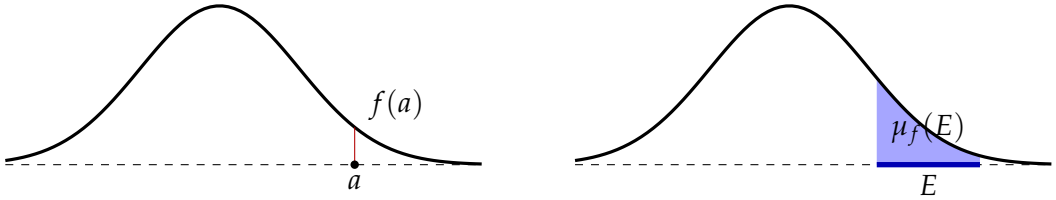


Figure 1.1: The same curve, viewed as a function (left) and as a measure (right).

The curve is the same; only the point of view has changed. Knowing the masses $\mu_f(E)$ for all Borel E is enough to recover f (almost everywhere), so nothing has been lost, and pointwise values can be discarded altogether. It is an abuse of language to say "functions *are* measures", but it is a productive one: we simply identify f with μ_f.

1.2.2 Not every measure is a function

The Dirac mass

The ***Dirac mass at*** 0 is the measure

$$\delta_0(E) := \begin{cases} 1, & 0 \in E, \\ 0, & \text{otherwise.} \end{cases}$$

Physicists call this the "delta function" and imagine a function that is zero away from the origin, with an infinitely tall spike at 0 enclosing total area one. No such function exists in the Lebesgue sense: a function vanishing off a single point has integral zero. The Dirac mass therefore lives in a strictly larger world.

Proposition 1.2.2

There is no $f \in L^1_{\mathrm{loc}}(\mathbb{R}^N)$ with $\delta_0(E) = \int_E f \, \mathrm{d}x$ for every Borel set E.

Proof. Take $E = \{0\}$. The left side equals 1, while the right side is an integral over a Lebesgue-null set and hence equals 0. □

1.2.3 When is a measure a function?

Suppose a measure μ happens to equal μ_f. By (1.1),

$$f(x) = \lim_{r\to 0^+} \frac{1}{|B(x,r)|}\int_{B(x,r)} f = \lim_{r\to 0^+} \frac{\mu(B(x,r))}{|B(x,r)|}$$

for almost every x. So *if* μ is a function, that function is forced to be the limiting density of μ. This suggests a representation theorem, and indeed one holds. When the limit exists it is called the ***Radon–Nikodym derivative*** of μ with respect to Lebesgue measure:

$$\frac{\mathrm{d}\mu}{\mathrm{d}x}(x) := \lim_{r\to 0^+} \frac{\mu(B(x,r))}{|B(x,r)|}.$$

Theorem 1.2.3 (When a measure is a function)
For a Radon measure μ on $\mathbb{R}^N$ the following are equivalent.

(a) $\mu = \mu_f$ for some $f \in L^1_{\mathrm{loc}}$ ("μ is a function").

(b) The limit $\lim_{r\to 0^+} \mu(B(x,r))/|B(x,r)|$ exists (and μ is recovered from it).

(c) The Radon–Nikodym derivative $\mathrm{d}\mu/\mathrm{d}x$ exists and represents μ.

(d) μ is ***absolutely continuous***: $\mu(E) = 0$ whenever $|E| = 0$. We write $\mu \ll \mathcal{L}^N$.

(e) Locally, for every $\varepsilon > 0$ there is $\delta > 0$ with $|E| < \delta \implies |\mu(E)| < \varepsilon$.

Condition (d) is the most useful: it says that μ is "compatible" with Lebesgue measure.

Example 1.2.4
A few instances of Theorem 1.2.3:

- The zero measure kills every null set, so it is a function — namely 0.
- Lebesgue measure is trivially compatible with itself; it is the function 1, since $|E| = \int_E 1\,\mathrm{d}x$.
- $\delta_0(\{0\}) = 1$ although $|\{0\}| = 0$; this incompatibility is exactly why δ_0 is not a function.
- Let F be a smooth vector field on $\mathbb{R}^2$ and, for bounded open connected U, set $\mu(U) := \oint_{\partial U} F \cdot \mathrm{d}r$ (taken counter-clockwise). Measure theory shows μ extends to a signed Radon measure; regions of small area have small circulation, so we expect $\mu \ll \mathcal{L}^2$. Indeed μ is the function $\operatorname{curl} F$ — this is Green's theorem.

1.3 The third layer: distributions

In the third layer one usually works in one of two regimes:

- the ***general case***, whose objects are simply called *distributions*; or
- the ***Fourier case***, whose objects are *tempered distributions* — distributions that interact well with the Fourier transform. Tempered distributions form a proper subclass of distributions.

At the second layer we viewed a function as something that assigns mass to sets. At the third we view it as something that *acts on other functions*. The functions it acts on are extremely well behaved and are called ***test functions***. Think of a test function as a small smooth bump which vanishes, or at least decays extremely fast, as $|x| \to \infty$. A standard example, valid in both regimes, is

$$\phi(x) = \begin{cases} e^{1/(x^2-1)}, & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

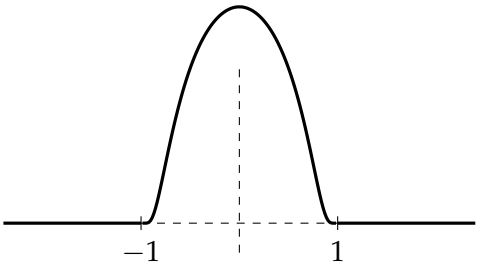


Figure 1.2: A smooth, compactly supported bump function (vertically rescaled).

To *test* a function f against ϕ means to form

$$\int_{\mathbb{R}^N} f(x)\phi(x)\,\mathrm{d}x.$$

If we know this number for every test function, we should be able to recover f without ever evaluating it at a point. This defines a linear map

$$T_f(\phi) := \int_{\mathbb{R}^N} f\phi$$

on the space of test functions. Since each f produces such a map, linear maps $T : \{\text{test functions}\} \to \mathbb{R}$ generalize functions, and these are essentially what distributions are. In analysis a linear map is of little use unless it is also *continuous*, which leads to the following.

Distribution, rough version
A ***distribution*** is a continuous linear map from test functions to numbers.

This is only a sketch until we specify which functions are "test functions" and what "continuous" means, i.e. which topology the space of test functions carries. The two regimes answer these questions differently.

1.3.1 The general case

Distributions can be defined on any open set $U \subseteq \mathbb{R}^N$.

Test functions
The space of test functions on U is $C_c^\infty(U)$, the set of smooth functions $U \to \mathbb{R}$ (or $\mathbb{C}$) whose support is a compact subset of U.

The natural topology on $C_c^\infty(U)$ is not metrizable, and its cleanest description is categorical (an inductive limit). The idea is sane, though. Suprema over open sets are badly behaved, while suprema over compact sets are not. So one first considers, for compact $K \subseteq U$, smooth functions supported in K, with the seminorms

$$\|\phi\|_{C^j(K)} := \sum_{|\alpha| \le j} \sup_K |\partial^\alpha \phi|, \qquad j = 0, 1, 2, \dots$$

These make each such space a (Fréchet) locally convex space. The topology on $C_c^\infty(U)$ is then the "limit" of these topologies as K grows to exhaust U.

Remark 1.3.1 (For reference only). Concretely: $\phi_n \to \phi$ in $C_c^\infty(U)$ iff there is a single compact K containing the supports of all ϕ_n and $\|\phi_n - \phi\|_{C^j(K)} \to 0$ for every j. A set $\Omega \subseteq C_c^\infty(U)$ is open iff for every compact K its trace on functions supported in K is open for the $C^\infty(K)$ topology. We will not use this directly.

Remark 1.3.2. It is convenient to allow complex-valued test functions and distributions, especially when comparing with the Fourier case. We will do so tacitly whenever needed.

Distribution
A ***distribution*** on $U \subseteq \mathbb{R}^N$ is a continuous linear map $T : C_c^\infty(U) \to \mathbb{R}$ (or $\mathbb{C}$). The space of distributions is denoted $\mathcal{D}'(U)$.

Some authors call these "Schwartz distributions"; we reserve Schwartz's name for the Fourier setting. In practice continuity is verified with the following criterion.

Theorem 1.3.3 (Continuity criterion for distributions)
A linear map $T : C_c^\infty(U) \to \mathbb{R}$ is continuous iff for every compact $K \subseteq U$ there exist $j \ge 0$ and $C_K > 0$ with

$$|T(\phi)| \le C_K \, \|\phi\|_{C^j(K)} \quad \text{for all } \phi \in C_c^\infty(U) \text{ with } \operatorname{supp}\phi \subseteq K.$$

The integer j and the constant C_K may depend on K. It does not matter whether one sums or takes the maximum over $|\alpha| \le j$ in the seminorm, since all norms on a finite-dimensional space are equivalent. From now on, unless stated otherwise, $U = \mathbb{R}^N$.

1.3.2 The Fourier case

Tempered distributions only make sense on all of $\mathbb{R}^N$, since that is where the Fourier transform lives.

Schwartz functions
A ***Schwartz function*** is a smooth $\phi : \mathbb{R}^N \to \mathbb{C}$ such that $|x^\alpha \, \partial^\beta \phi(x)|$ is bounded for all multi-indices α, β. The space of Schwartz functions is $\mathcal{S}(\mathbb{R}^N)$.

In words: ϕ and all its derivatives decay faster than every polynomial. Any smooth function whose derivatives all decay exponentially is Schwartz.

Example 1.3.4
Clearly $C_c^\infty(\mathbb{R}^N) \subseteq \mathcal{S}(\mathbb{R}^N)$, since a function that eventually vanishes certainly decays rapidly. The inclusion is strict: the Gaussian $\phi(x) = e^{-|x|^2}$ is Schwartz but not compactly supported.

So in the specialized Fourier regime we test against *more* functions than in general. Passing more tests is a stronger requirement, which is why tempered distributions will form a *smaller* class.

The topology on $\mathcal{S}(\mathbb{R}^N)$ is far simpler: it is the locally convex topology generated by the seminorms

$$\|\phi\|_{m,n} := \sup_{|\alpha|\le m} \sup_{|\beta|\le n} \sup_{x\in\mathbb{R}^N} |x^\alpha \partial^\beta \phi(x)|, \qquad m,n \ge 0.$$

Equivalently, $\phi_k \to \phi$ in $\mathcal{S}$ iff $x^\alpha \partial^\beta \phi_k \to x^\alpha \partial^\beta \phi$ uniformly on $\mathbb{R}^N$ for every α, β.

Tempered distribution
A ***tempered distribution*** is a continuous linear map $T : \mathcal{S}(\mathbb{R}^N) \to \mathbb{C}$. The space of tempered distributions is denoted $\mathcal{S}'(\mathbb{R}^N)$.

Theorem 1.3.5 (Continuity criterion for tempered distributions)
A linear map $T : \mathcal{S}(\mathbb{R}^N) \to \mathbb{C}$ is continuous iff there exist $m, n \ge 0$ and $C > 0$ with $|T(\phi)| \le C\|\phi\|_{m,n}$ for all $\phi \in \mathcal{S}(\mathbb{R}^N)$.

Compared with Theorem 1.3.3, a *single* estimate suffices. For instance, if $|T(\phi)| \le C\|\nabla\phi\|_\infty$ then T is tempered (take $m = 0, n = 1$).

1.3.3 Tempered distributions are distributions

Proposition 1.3.6
If T is a tempered distribution, then its restriction to $C_c^\infty(\mathbb{R}^N)$ is a distribution. Thus, abusing notation, $\mathcal{S}'(\mathbb{R}^N) \subseteq \mathcal{D}'(\mathbb{R}^N)$.

Proof. Let $|T(\phi)| \le C\|\phi\|_{m,n}$. If $\operatorname{supp}\phi \subseteq K \subseteq B(0,R)$ then $|x^\alpha \partial^\beta \phi| \le (1+R)^m \sup_K |\partial^\beta \phi|$ for $|\alpha| \le m$, so $|T(\phi)| \le C(1+R)^m \|\phi\|_{C^n(K)}$, which is the criterion of Theorem 1.3.3. □

So distributions are more general and can represent more objects, while tempered distributions are specialized, deliberately, so as to cooperate with the Fourier transform. The reason for testing against the larger space $\mathcal{S}$ is the following fact.

Theorem 1.3.7
The Fourier transform is a continuous linear bijection $\mathcal{F} : \mathcal{S}(\mathbb{R}^N) \to \mathcal{S}(\mathbb{R}^N)$ with continuous inverse $\mathcal{F}^{-1}$.

By contrast, $\mathcal{F}$ does *not* map C_c^∞ into C_c^∞: a nonzero function and its Fourier transform can never both be compactly supported (a form of the uncertainty principle). The space $\mathcal{S}$ is exactly the right size.

1.3.4 Functions and measures are distributions

Given a function f on $\mathbb{R}^N$, "regarding f as a distribution" always means identifying f with the functional

$$T_f(\phi) := \int_{\mathbb{R}^N} f\phi,$$

with ϕ ranging over whichever class of test functions is in play. One can show that distinct functions give distinct functionals, so this identification is legitimate.

Theorem 1.3.8
Every $f \in L^1_{\text{loc}}(\mathbb{R}^N)$ is a distribution: $T_f : C_c^\infty(\mathbb{R}^N) \to \mathbb{R}$ is linear and continuous. In particular every L^p function is a distribution.

Fewer functions are tempered. There is no clean characterization, but the following class is large.

Theorem 1.3.9
If $f = Pg$ with $g \in L^p(\mathbb{R}^N)$ for some $1 \le p \le \infty$ and P a polynomial, then $T_f : \mathcal{S}(\mathbb{R}^N) \to \mathbb{C}$ is a tempered distribution.

Measures are distributions too. A measure μ is identified with

$$T_\mu(\phi) := \int_{\mathbb{R}^N} \phi \, \mathrm{d}\mu.$$

This is consistent with the identification of f with μ_f: since $\mathrm{d}\mu_f = f \, \mathrm{d}x$, we have $\int \phi \, \mathrm{d}\mu_f = \int \phi f \, \mathrm{d}x$, so f and μ_f define the same distribution.

Theorem 1.3.10
Every (signed or unsigned) Radon measure is a distribution.

Theorem 1.3.11
If a (signed or unsigned) Radon measure μ satisfies $|\mu|(B(0,R)) \le A(1+R^B)$ for all $R > 0$ and some $A, B > 0$, then μ is a tempered distribution.

The growth bound is sufficient but not necessary. Since an L^1_{loc} function is a Radon measure, Theorem 1.3.10 recovers Theorem 1.3.8; similarly, Theorem 1.3.11 implies Theorem 1.3.9 (a pleasant exercise with Hölder's inequality).

2 Distributions, Measures, and First Examples

2.1 The bracket notation

So far we have written $T(\phi)$ for a distribution applied to a test function. That is the literally correct notation, but it hides structure. For a function f we have $T_f(\phi) = \int f\phi$, which is an inner product of f and ϕ. Since we identify f with T_f, it is natural to write

$$\langle T, \phi \rangle := T(\phi)$$

for any distribution T. Likewise, for a measure, $\langle \mu, \phi \rangle := \int \phi \, d\mu$.

The payoff is immediate. We will define the derivative of a distribution on $\mathbb{R}$ by

$$\langle T', \phi \rangle := - \langle T, \phi' \rangle ,$$

which reads as "integration by parts", whereas $T'(\phi) := -T(\phi')$ looks like a typo.

> *Remark 2.1.1.* Some authors write $\langle \phi, T \rangle$ instead. We always put the distribution first.

2.2 When is a distribution a measure?

We now ask, one layer up, the same question we asked of measures.

2.2.1 Not every distribution is a measure

Consider the distribution on $\mathbb{R}$ that evaluates the derivative at the origin:

$$\langle T, \phi \rangle := \phi'(0).$$

There is no measure μ with $\int \phi \, d\mu = \phi'(0)$ for all ϕ: integrals against a measure cannot "see" derivatives. One can perturb ϕ by a tiny amount in sup norm while making $\phi'(0)$ enormous. (This is made precise in Example 2.3.4 below.)

2.2.2 Positive distributions

It turns out to be easier to first characterize *unsigned* measures.

Theorem 2.2.1 (Positive distributions are measures)
A distribution (or tempered distribution) T is an unsigned Radon measure iff it is ***positive***: $\langle T, \phi \rangle \geq 0$ for every test function $\phi \geq 0$.

This is a close relative of the Riesz representation theorem, which says that positive linear functionals on *continuous* compactly supported functions are Radon measures. Testing only against smooth functions is a weaker demand, so a priori it could allow more objects; the theorem says it does not.

Theorem 2.2.2
A distribution (or tempered distribution) is a signed Radon measure iff it is a difference of two positive distributions.

2.2.3 The order of a distribution

A more practical criterion uses the integer j of Theorem 1.3.3.

Order
The ***order*** of a distribution T is the smallest j (if one exists) such that for every compact K there is C_K with $|\langle T, \phi\rangle| \le C_K \|\phi\|_{C^j(K)}$ for all ϕ supported in K.

Intuitively, the order is the number of derivatives of ϕ that T needs in order to compute $\langle T, \phi\rangle$. For instance, an estimate $|\langle T, \phi\rangle| \le C_K(\|\phi\|_{C^0(K)} + \|\phi'\|_{C^0(K)})$ shows that T has order at most 1. Since measures are blind to derivatives, we expect measures to have order zero, and remarkably this characterizes them.

Theorem 2.2.3 (Order zero means measure)
A distribution T is a Radon measure iff it has order 0, that is, for every compact K there is C_K with

$$|\langle T, \phi\rangle| \le C_K \sup_K |\phi| \quad \text{for all } \phi \text{ supported in } K.$$

For a tempered distribution, the order is the least n for which an estimate $|\langle T, \phi\rangle| \le C\|\phi\|_{m,n}$ holds for some m.

Theorem 2.2.4
A tempered distribution T is a Radon measure iff it has order 0, i.e. for some m and C,

$$|\langle T, \phi\rangle| \le C \sup_{|\alpha|\le m} \sup_{x\in\mathbb{R}^N} |x^\alpha \phi(x)| \quad \text{for all } \phi \in \mathcal{S}(\mathbb{R}^N).$$

2.2.4 When is a distribution a function?

Chaining Theorem 2.2.3 with Theorem 1.2.3 decides which distributions are functions: T is a function iff it has order 0 and the resulting measure is absolutely continuous. There does not seem to be a more direct criterion.

2.3 First examples

Example 2.3.1 (Functions as distributions)
Let $f \in L^1_{\text{loc}}$ and $\langle T_f, \phi\rangle := \int f\phi \, dx$. We already know T_f is a distribution, but it is instructive to run the criteria.

- For ϕ supported in a compact K, $|\langle T_f, \phi\rangle| \le \int_K |f\phi| \le \|f\|_{L^1(K)} \|\phi\|_{C^0(K)}$, so T_f is a distribution of order 0, hence a measure (namely μ_f).
- The same idea fails for Theorem 1.3.5: T_f need not be tempered. The function e^{x^2} on $\mathbb{R}$ is a distribution, but not a tempered one; its pairing with the Schwartz function $e^{-x^2/2}$ already diverges.
- If $f \in L^1$, then $|\langle T_f, \phi\rangle| \le \|f\|_{L^1}\|\phi\|_{L^\infty} = \|f\|_{L^1}\|\phi\|_{0,0}$, so T_f is tempered.

- If $f \in L^p$, choose m so large that $(1+|x|^m)^{-1} \in L^{p'}$. Then by Hölder,

$$\begin{aligned} |\langle T_f, \phi\rangle| &\le \int |f| \frac{(1+|x|^m)|\phi|}{1+|x|^m}\,\mathrm{d}x \\ &\le \|f\|_{L^p} \left\|(1+|x|^m)^{-1}\right\|_{L^{p'}} \sup_x (1+|x|^m)|\phi(x)| \le C\|\phi\|_{m,0}. \end{aligned}$$

  So every L^p function is tempered, and one may (abusively) write $\mathcal{S}(\mathbb{R}^N) \subseteq L^p(\mathbb{R}^N) \subseteq \mathcal{S}'(\mathbb{R}^N)$.

Example 2.3.2 (Measures as distributions)
Let μ be Radon and $\langle T_\mu, \phi\rangle := \int \phi\,\mathrm{d}\mu$.

- For ϕ supported in compact K, $|\langle T_\mu, \phi\rangle| \le \|\phi\|_{C^0(K)}\,|\mu|(K)$, and $|\mu|(K) < \infty$ because μ is Radon. So T_μ is a distribution of order 0.
- To be tempered we need control at infinity. If $|\mu|(B(0,R)) \le A(1+R^k)$, then

$$|\langle T_\mu, \phi\rangle| \le \int |\phi|\,(1+|x|^{k+N+1})\,\frac{\mathrm{d}|\mu|}{1+|x|^{k+N+1}} \le C\|\phi\|_{k+N+1,0} \int \frac{\mathrm{d}|\mu|}{1+|x|^{k+N+1}},$$

  and the last integral is finite: decompose $\mathbb{R}^N$ into dyadic shells and use the growth bound on each.

Example 2.3.3 (The Dirac delta)
The ***delta distribution*** at 0 is the evaluation map

$$\langle \delta_0, \phi\rangle := \phi(0).$$

- It is clearly linear. If $\operatorname{supp}\phi \subseteq K$ then $|\langle \delta_0, \phi\rangle| \le \sup_K |\phi| = \|\phi\|_{C^0(K)}$ (trivially if $0 \notin K$). So δ_0 is a distribution of order 0.
- It is tempered: $|\phi(0)| \le \sup|\phi| = \|\phi\|_{0,0}$.
- Being of order 0, it is a measure by Theorem 2.2.3. Alternatively, $\phi \ge 0$ implies $\phi(0) \ge 0$, so δ_0 is positive and Theorem 2.2.1 applies. The measure is, of course, the Dirac mass: it gives E mass 1 if $0 \in E$ and 0 otherwise.
- Since $\delta_0(\{0\}) = 1$ but $|\{0\}| = 0$, Theorem 1.2.3 shows that δ_0 is not a function.

More generally, $\langle \delta_a, \phi\rangle := \phi(a)$ defines the ***delta at*** a, the measure assigning mass 1 to sets containing a. It is common to write δ for δ_0 and $\delta(x-a)$ for δ_a.

Example 2.3.4 (Evaluating a derivative)
On $\mathbb{R}$, let $\langle T, \phi\rangle := \phi'(0)$.

- If $\operatorname{supp}\phi \subseteq K$, then $|\langle T, \phi\rangle| \le \|\phi\|_{C^1(K)}$, so T is a distribution of order at most 1.
- It is tempered, since $|\phi'(0)| \le \|\phi\|_{0,1}$.
- It is not a measure. Otherwise, taking $K = [-1,1]$ in Theorem 2.2.3, there would

be C with $|\phi'(0)| \le C \sup_{[-1,1]} |\phi|$ for all ϕ supported in $[-1,1]$. But we can choose such ϕ with $\sup|\phi| \le 10^{-5}$ and $\phi'(0) = 10^5 C$ (a tiny, steep bump), a contradiction.

Example 2.3.5 (The principal value of $1/x$)
The function $1/x$ is not locally integrable near 0, so it does not define a distribution by integration. The standard repair is the ***principal value***

$$\langle \mathrm{p.v.}\, \tfrac{1}{x}, \phi \rangle := \lim_{\varepsilon \to 0^+} \int_{|x|>\varepsilon} \frac{\phi(x)}{x}\, \mathrm{d}x.$$

That the limit exists, and defines a tempered distribution, follows from one decomposition. Since $1/x$ is odd, $\int_{\varepsilon<|x|<1} \phi(0)/x\, \mathrm{d}x = 0$, so

$$\langle \mathrm{p.v.}\, \tfrac{1}{x}, \phi \rangle = \int_{|x|\ge 1} \frac{x\phi(x)}{x^2}\, \mathrm{d}x + \lim_{\varepsilon \to 0^+} \int_{\varepsilon<|x|<1} \frac{\phi(x)-\phi(0)}{x}\, \mathrm{d}x.$$

By the mean value theorem the second integrand is bounded by $\sup|\phi'|$, so the limit exists. Estimating both terms,

$$\left| \langle \mathrm{p.v.}\, \tfrac{1}{x}, \phi \rangle \right| \le 2 \sup_x |x\phi(x)| + 2 \sup_x |\phi'(x)| \le 4 \|\phi\|_{1,1}.$$

Hence p.v. $\frac{1}{x}$ is tempered, of order at most 1. Its order is in fact exactly 1 (try to prove it), so it is not a measure.

More generally, if $g \in L^1_{\mathrm{loc}}(\mathbb{R}^N \setminus B(0,\delta))$ for every $\delta > 0$ and g is tame enough near 0, one can define

$$\langle \mathrm{p.v.}\, g, \phi \rangle := \lim_{\delta \to 0^+} \int_{\mathbb{R}^N \setminus B(0,\delta)} g\phi\, \mathrm{d}x.$$

Example 2.3.6 (A preview: convolution of measures)
For finite measures μ, ν on $\mathbb{R}^N$ consider

$$\langle \mu * \nu, \phi \rangle := \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \phi(x+y)\, \mathrm{d}\mu(x)\, \mathrm{d}\nu(y).$$

(This is not the general definition of convolution, which comes later, but it is correct for finite measures.) It is easy to check that this is a distribution. But notice: the right side involves no derivatives of ϕ, and $|\langle \mu * \nu, \phi \rangle| \le \sup|\phi| \cdot |\mu|(\mathbb{R}^N)|\nu|(\mathbb{R}^N)$. So $\mu * \nu$ has order 0 and must be a measure! Which measure? If it is a measure, then $\int \phi\, \mathrm{d}(\mu*\nu) = \iint \phi(x+y)\, \mathrm{d}\mu\, \mathrm{d}\nu$ must also hold for indicator functions $\phi = \mathbf{1}_E$, whence

$$(\mu * \nu)(E) = \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \mathbf{1}_E(x+y)\, \mathrm{d}\mu(x)\, \mathrm{d}\nu(y).$$

With no computation at all, the theory has told us both that $\mu * \nu$ is a measure and what its formula must be.

3 Operations on Distributions

3.1 Addition

If S and T are distributions, so is $S+T$, defined by $\langle S+T,\,\phi\rangle := \langle S,\,\phi\rangle + \langle T,\,\phi\rangle$. Scalar multiples are defined the same way. Nothing more needs saying.

3.2 Multiplication, and why it is hard

For distributions S, T the map $\phi \mapsto \langle S,\,\phi\rangle\,\langle T,\,\phi\rangle$ is usually not even linear, so it is certainly not "the product ST". There is no reasonable meaning for expressions such as δ_0^2. What we *can* do is multiply by smooth functions.

Multiplication by a smooth function
Let T be a distribution and $f \in C^\infty(\mathbb{R}^N)$. The distribution fT is

$$\langle fT,\,\phi\rangle := \langle T,\,f\phi\rangle\,.$$

This makes sense because $f\phi \in C_c^\infty$ whenever $\phi \in C_c^\infty$; one checks with the Leibniz rule that fT is again continuous.

⚠ In the tempered world more care is needed: if T is tempered, fT need not be. Take $T = 1$ and $f(x) = e^{|x|^2}$; then fT is the function $e^{|x|^2}$, which is not tempered. For fT to be tempered one requires, for instance, that f and all of its derivatives grow at most polynomially.

3.3 Division, and why not to try (yet)

On $\mathbb{R}$, which distributions T satisfy $xT = 0$? "Dividing by x" suggests $T = 0$, and $T = 0$ is indeed a solution, but it is not the only one: the point $x = 0$, where we cannot divide, causes trouble. We will return to this in Chapter 4, after we have derivatives available.

3.4 Differentiation

Differentiation is the heart of the matter — perhaps ninety percent of the reason distributions exist. For $f \in C^1(\mathbb{R})$ and $\phi \in C_c^\infty(\mathbb{R})$, integration by parts gives $\int f'\phi = -\int f\phi'$, i.e. $\langle f',\,\phi\rangle = -\langle f,\,\phi'\rangle$. We simply declare this to be the definition for every distribution.

Distributional derivative
For a distribution T on $\mathbb{R}^N$, the partial derivative $\partial_{x_j} T$ is the distribution

$$\left\langle \partial_{x_j} T,\,\phi\right\rangle := -\left\langle T,\,\partial_{x_j}\phi\right\rangle.$$

More generally, for every multi-index α,

$$\langle \partial^\alpha T,\,\phi\rangle := (-1)^{|\alpha|}\,\langle T,\,\partial^\alpha\phi\rangle\,.$$

The burden of differentiation is thrown entirely onto the test function, which can bear any number of derivatives. The following facts are standard.

Proposition 3.4.1
Let T be a distribution.

- If T is a distribution, so is $\partial^\alpha T$; if T is tempered, so is $\partial^\alpha T$.
- Differentiation is linear.
- (Product rule) For f smooth, $\partial_{x_j}(fT) = (\partial_{x_j} f)T + f\,\partial_{x_j} T$.

3.4.1 Differentiation loses structure

Taking derivatives generally degrades a distribution. Consider

$$f(x) = \begin{cases} 5, & x < 0, \\ 6, & x \geq 0. \end{cases}$$

This is a function, hence a distribution, hence differentiable. We will see (Example 3.7.3) that $f' = \delta_0$: differentiation has turned a function into a measure that is not a function. Differentiating again gives $f'' = \delta_0'$, which is not even a measure (it is, up to sign, the distribution of Example 2.3.4).

Conversely, if the derivatives of T are well structured, so is T. Two important classes of functions are defined this way.

- If the first derivatives of a function f are *measures*, f has ***bounded variation***; we write $f \in BV$ (see Section 3.7.4).
- If the first derivatives of f are *functions*, f is ***Sobolev***; under integrability assumptions it lies in a Sobolev space $W^{1,p}$ (see Section 3.7.3). These derivatives are called ***weak derivatives***.

3.5 Adjoints

A small piece of notation will save much pain later. In general, the ***adjoint*** of an operator A is the operator A^* satisfying $\langle Ax,\, y\rangle = \langle x,\, A^* y\rangle$; in linear algebra this is the (conjugate) transpose. In our setting, if A acts on test functions, then A^* acts on distributions and

$$\langle T,\, A\phi\rangle = \langle A^* T,\, \phi\rangle\,,$$

and symmetrically if A acts on distributions.

Differential operators act on both. For example, if T is a distribution on $\mathbb{R}^2$, then

$$\Big\langle (\partial_x + \partial_y^2)T,\, \phi \Big\rangle = \Big\langle T,\, (-\partial_x + \partial_y^2)\phi \Big\rangle\,,$$

so the adjoint of $\partial_x + \partial_y^2$ is $-\partial_x + \partial_y^2$. A differential operator is generally not its own adjoint, so we cannot write $\langle T,\, D\phi\rangle = \langle DT,\, \phi\rangle$. We can, however, always write

$$\langle T,\, D\phi\rangle = \langle D^* T,\, \phi\rangle\,, \qquad \text{where} \quad D = \sum_{|\alpha|\leq n} c_\alpha \partial^\alpha \implies D^* = \sum_{|\alpha|\leq n} (-1)^{|\alpha|} c_\alpha \partial^\alpha,$$

and the formula on the right need never be written again.

3.6 The structure theorem

The motivation for distributions was to differentiate the non-differentiable. Remarkably, that is also the *only* way distributions arise.

Theorem 3.6.1 (Local structure of distributions)
Let T be a distribution on $\mathbb{R}^N$.

- Locally, T is a derivative of a continuous function: for every compact K there are a continuous f and a multi-index α with $\langle T, \phi\rangle = (-1)^{|\alpha|}\int f\,\partial^\alpha\phi\,\mathrm{d}x$ for all ϕ supported in K.
- Globally, $T = \sum_\beta \partial^\beta f_\beta$, a locally finite sum of derivatives of continuous functions f_β.

Heuristically: differentiation can only increase the order of a distribution, so repeatedly taking *antiderivatives* should decrease it until we reach a continuous function. This works on compact sets; a partition of unity then glues the local pictures together, which accounts for the sum. We will not use this theorem, but it is too illuminating to omit.

3.7 Examples

We work in one dimension unless stated otherwise.

Example 3.7.1 (Classical derivatives)
Let $f \in C^1(\mathbb{R})$ and let T be its distributional derivative. Then

$$\langle T, \phi\rangle = -\langle f, \phi'\rangle = -\int f\phi' = \int f'\phi = \langle f', \phi\rangle.$$

So T is the function f': distributional differentiation agrees with the classical one.

Example 3.7.2 (The absolute value)
Calculus says $|x|$ is not differentiable. Its distributional derivative is

$$-\int_{\mathbb{R}} |x|\phi'(x)\,\mathrm{d}x = -\int_0^\infty x\phi'\,\mathrm{d}x + \int_{-\infty}^0 x\phi'\,\mathrm{d}x = \int_0^\infty \phi - \int_{-\infty}^0 \phi = \int (\operatorname{sgn} x)\phi,$$

after integrating by parts on each half-line. Hence

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}x}|x| = \operatorname{sgn} x}$$

The derivative is a function, so $|x|$ is (locally) Sobolev.

Example 3.7.3 (A jump discontinuity)
Let $f = a$ on $(-\infty, 0)$ and $f = b$ on $[0, \infty)$. Then

$$\langle f', \phi\rangle = -\int f\phi' = -a\int_{-\infty}^0 \phi' - b\int_0^\infty \phi' = -a\phi(0) + b\phi(0),$$

so $f' = (b-a)\,\delta_0$: the derivative of a jump is a delta whose weight is the size of the jump. In particular the ***Heaviside function*** $H = \mathbf{1}_{[0,\infty)}$ satisfies $H' = \delta_0$, and $\operatorname{sgn}' = 2\delta_0$. Since f' is a measure, f has bounded variation.

Example 3.7.4 (Derivatives of the delta)
In $\mathbb{R}^N$, directly from the definitions,

$$\langle \partial^\alpha \delta_0,\, \phi\rangle = (-1)^{|\alpha|}\,\langle \delta_0,\, \partial^\alpha \phi\rangle = (-1)^{|\alpha|}\partial^\alpha\phi(0).$$

In one dimension, $\left\langle \delta_0^{(k)},\, \phi \right\rangle = (-1)^k \phi^{(k)}(0)$; we also write $\partial_x^k \delta_0$. The same holds with 0 replaced by any point a.

Example 3.7.5 (Lebesgue measure)
Purely for fun, differentiate one-dimensional Lebesgue measure $\mathcal{L}^1$:

$$\left\langle (\mathcal{L}^1)',\, \phi \right\rangle = -\int \phi'\,\mathrm{d}\mathcal{L}^1 = -\int \phi'\,\mathrm{d}x = 0.$$

This is no surprise once one recalls that Lebesgue measure is the function 1.

Proposition 3.7.6 (Vanishing derivative)
On $\mathbb{R}$, $T' = 0$ iff T is a constant function.

Proof. Constants clearly have zero derivative. Conversely suppose $T' = 0$. Fix $\theta \in C_c^\infty(\mathbb{R})$ with $\int \theta = 1$. Given $\phi \in C_c^\infty$, the function $\phi - (\int \phi)\theta$ has integral zero, so its antiderivative $\psi(x) := \int_{-\infty}^{x} \left(\phi - (\int \phi)\theta\right)$ is again in C_c^∞, with $\psi' = \phi - (\int \phi)\theta$. Therefore

$$\langle T,\, \phi\rangle = \left(\int \phi\right)\langle T,\, \theta\rangle + \langle T,\, \psi'\rangle = c\int \phi - \langle T',\, \psi\rangle = \langle c,\, \phi\rangle\,, \qquad c := \langle T,\, \theta\rangle\,. \quad \square$$

The same is true in $\mathbb{R}^N$: if $\nabla T = 0$ (every $\partial_{x_j} T = 0$) then T is constant.

Example 3.7.7 (The logarithm)
The function $\log|x|$ is locally integrable, so it is a distribution, and its pairing can be computed as a principal value $\lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \log|x|\,\phi\,\mathrm{d}x$ (the logarithmic singularity is integrable, so no principal value is needed). We claim

$$(\log|x|)' = \mathrm{p.v.}\,\frac{1}{x}.$$

Integrating by parts on (ε,∞) and on $(-\infty,-\varepsilon)$,

$$-\int_{|x|>\varepsilon} \log|x|\,\phi'(x)\,\mathrm{d}x = \int_{|x|>\varepsilon} \frac{\phi(x)}{x}\,\mathrm{d}x + \log\varepsilon\,\big(\phi(\varepsilon) - \phi(-\varepsilon)\big).$$

The boundary term is $O(\varepsilon\log\varepsilon) \to 0$, and letting $\varepsilon \to 0$ proves the claim.

3.7.1 Monotone functions and Lebesgue–Stieltjes measures

Let $f : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous. By Lebesgue's theorem f is differentiable almost everywhere, but its pointwise derivative does *not* capture f: the Cantor function (the "devil's staircase") is increasing, continuous and non-constant, yet has derivative 0 almost everywhere. The distributional derivative gets it right.

Proposition 3.7.8
If f is increasing and right-continuous, then f' is the ***Lebesgue–Stieltjes measure*** of f, i.e. the unique measure μ with

$$f(b) = f(a) + \mu((a,b]) = f(a) + \int_{(a,b]} \mathrm{d}\mu \qquad (a < b).$$

Compare the fundamental theorem of calculus, $g(b) = g(a) + \int_a^b g' \,\mathrm{d}x$.

Proof. Let $\operatorname{supp}\phi \subseteq (a,b)$. Writing $f(x) = f(a) + \int \mathbf{1}_{(a,x]}(y)\,\mathrm{d}\mu(y)$ for $x \in (a,b)$ and using Fubini,

$$\begin{aligned} -\int_a^b f\phi' \,\mathrm{d}x &= -f(a)\int_a^b \phi' - \int_{(a,b]} \int_y^b \phi'(x)\,\mathrm{d}x\,\mathrm{d}\mu(y) \\ &= 0 - \int_{(a,b]} \big(\phi(b) - \phi(y)\big)\,\mathrm{d}\mu(y) = \int \phi \,\mathrm{d}\mu. \end{aligned}$$

□

The statement and proof extend verbatim to right-continuous functions of locally bounded (pointwise) variation. In probability, a random variable X has a law μ_X and a right-continuous CDF F_X, and μ_X is precisely the Lebesgue–Stieltjes measure of F_X. So $F_X' = \mu_X$ in the sense of distributions, and when this distribution is a function it is the density f_X.

3.7.2 Absolutely continuous functions

Proposition 3.7.9
Let $f : [a,b] \to \mathbb{R}$. Then the distributional derivative f' is a function iff f (has a representative which) is absolutely continuous.

This is why absolute continuity is so pleasant. For example $|x|$ is absolutely continuous, and indeed its derivative $\operatorname{sgn} x$ was a function.

3.7.3 Sobolev spaces

In $\mathbb{R}^N$, suppose f is a function whose first partial derivatives are all functions. If f and its first derivatives lie in L^p, we say $f \in W^{1,p}(\mathbb{R}^N)$. More generally, $f \in W^{k,p}$ if all distributional derivatives of f of order at most k are L^p functions. In one dimension, by the previous proposition, $W^{1,p}$ is easy to understand through absolute continuity.

Because distributions can be restricted to open sets, Sobolev spaces make sense on domains $\Omega \subseteq \mathbb{R}^N$. On all of $\mathbb{R}^N$ one also has tempered distributions, and the Fourier transform turns out to be intimately connected to Sobolev regularity — a beautiful topic that we leave to the references.

3.7.4 Functions of bounded variation

Proposition 3.7.10
On $\mathbb{R}$, f' is a Radon measure iff f has locally bounded ***essential variation***, i.e. some representative of f has locally bounded pointwise variation.

The word "essential" is needed because L^1_{loc} functions have no pointwise values. One defines the essential variation of f as $\inf\{\operatorname{Var} \tilde{f} : \tilde{f} \text{ a representative of } f\}$; it is a good exercise to show that the infimum is attained.

Sketch. ($\Leftarrow$) A right-continuous function of locally bounded variation is a difference of increasing right-continuous functions, whose derivatives are measures by the previous subsection. ($\Rightarrow$) If $f' = \mu$, let $g(x) := \mu((0, x])$ for $x \geq 0$ and $g(x) := -\mu((x, 0])$ for $x < 0$. Then $g' = \mu$ (by the same Fubini computation), so $(f - g)' = 0$ and $f = g + c$ almost everywhere by Proposition 3.7.6. Finally g has variation at most $|\mu|$ on each interval. □

In several variables the definition is simply the following.

BV
A function $f \in L^1(\Omega)$ belongs to $BV(\Omega)$ if its first partial derivatives are finite Radon measures on Ω.

By the proposition, $BV(\mathbb{R})$ contains every integrable function of bounded pointwise variation.

4 Simple Division Problems

Let f be smooth and g a distribution. How do we solve $fT = g$ for T? For functions this is easy: divide by f, taking some care where f vanishes. For distributions it is genuinely difficult, and surprisingly little expository material exists on the subject. This chapter treats the easy cases, which already suffice for many applications. The general problem is far harder.

4.1 The equation $xT = 0$

Stay in one dimension and take $f(x) = x$. Away from 0 we may "divide by x" and conclude that T vanishes there. At 0 we cannot divide, so T might carry some weight at the origin. This suggests:

Proposition 4.1.1
A distribution T on $\mathbb{R}$ satisfies $xT = 0$ iff $T = a\delta_0$ for some constant a.

Proof. If $T = a\delta_0$ then $\langle xT, \phi\rangle = a\,(x\phi)(0) = 0$. Conversely, suppose $xT = 0$. Fix $\chi \in C_c^\infty(\mathbb{R})$ with $\chi \equiv 1$ near 0. For any $\phi \in C_c^\infty$, the function $\phi - \phi(0)\chi$ vanishes at 0, so by Taylor's theorem (in Hadamard's form) it equals $x\psi(x)$ with $\psi(x) = \int_0^1 (\phi - \phi(0)\chi)'(tx)\,dt$, which is smooth and compactly supported. Hence

$$\langle T, \phi\rangle = \phi(0)\,\langle T, \chi\rangle + \langle T, x\psi\rangle = \phi(0)\,\langle T, \chi\rangle + \langle xT, \psi\rangle = a\,\phi(0),$$

with $a := \langle T, \chi\rangle$. So $T = a\delta_0$. □

If the right side is a smooth function f, we can solve $xT = f$ by noting that $x \cdot \mathrm{p.v.}\,\frac{1}{x} = 1$ (indeed $\langle \mathrm{p.v.}\,\frac{1}{x}, x\phi\rangle = \lim_\varepsilon \int_{|x|>\varepsilon} \phi = \int \phi$), so that $x\big(T - f \cdot \mathrm{p.v.}\,\frac{1}{x}\big) = 0$. Hence the general solution is $T = f \cdot \mathrm{p.v.}\,\frac{1}{x} + a\delta_0$. This trick fails when the right side is a distribution S, since $\mathrm{p.v.}\,\frac{1}{x} \cdot S$ would be a product of two distributions. Nevertheless the equation is always solvable:

Exercise 4.1.2. Let S be any distribution on $\mathbb{R}$ and fix χ as in the proof above. Show that

$$\langle T, \phi\rangle := \Big\langle S,\ \frac{\phi - \phi(0)\chi}{x}\Big\rangle$$

defines a distribution with $xT = S$, and that every solution of $xT = S$ is $T + a\delta_0$ for some constant a.

4.2 Support

Support of a distribution
Let T be a distribution.

- T ***vanishes*** on an open set U if $\langle T, \phi\rangle = 0$ for all ϕ supported in U.
- The union of all open sets on which T vanishes is the largest open set on which T vanishes.
- The ***support*** $\operatorname{supp} T$ is the complement of that set.

For example $\operatorname{supp}\delta_0 = \{0\}$, and for continuous functions this agrees with the usual notion of support. We are interested in distributions whose support is tiny — a point, or a discrete set — because of the following.

Proposition 4.2.1
If f is smooth and $fT = 0$, then $\operatorname{supp} T \subseteq \{f = 0\}$.

Proof. Let U be an open set on which $f \neq 0$ and let $\operatorname{supp}\phi \subseteq U$. Then ϕ/f is smooth and compactly supported in U, and $\langle T, \phi\rangle = \langle fT, \phi/f\rangle = 0$. So T vanishes on $\{f \neq 0\}$. □

⚠ The converse is false. With $f(x) = x$ and $T = \delta_0'$ we have $\operatorname{supp} T = \{0\} = \{f = 0\}$, yet $\langle x\delta_0', \phi\rangle = -(x\phi)'(0) = -\phi(0)$, so $xT = -\delta_0 \neq 0$.

Theorem 4.2.2 (Distributions supported at a point)
If $\operatorname{supp} T = \{a\} \subseteq \mathbb{R}^N$, then $T = D\delta_a$ for some linear differential operator $D = \sum_{|\alpha|\le j} c_\alpha \partial^\alpha$ with constant coefficients and finite order j. Conversely, every such $D\delta_a$ is supported in $\{a\}$.

A proof is in Rudin [3, Thm. 6.25]. In particular such a T has finite order; in fact every compactly supported distribution has finite order (an exercise with Theorem 1.3.3).

4.3 Zeros of finite order

The following computation is the engine for everything in this section.

Lemma 4.3.1
On $\mathbb{R}$, $(x-a)^k\,\delta_a^{(j)} = 0$ if $j < k$, while for $j \ge k$

$$\left\langle (x-a)^k \delta_a^{(j)}, \phi \right\rangle = (-1)^j \frac{j!}{(j-k)!}\,\phi^{(j-k)}(a).$$

Proof. $\left\langle (x-a)^k\delta_a^{(j)}, \phi\right\rangle = (-1)^j\big((x-a)^k\phi\big)^{(j)}(a)$. By the Leibniz rule, the only term that survives at $x = a$ is the one in which exactly k derivatives fall on $(x-a)^k$, contributing $k!\binom{j}{k}\phi^{(j-k)}(a)$ when $j \ge k$ and nothing otherwise. □

Proposition 4.3.2
Fix $a \in \mathbb{R}$ and $n \ge 1$. Then $(x-a)^n T = 0$ iff $T = \sum_{k=0}^{n-1} c_k \delta_a^{(k)}$ for constants $c_0, \dots, c_{n-1}$; that is, $T = D\delta_a$ with D of order at most $n-1$.

Proof. If T has this form, $(x-a)^n T = 0$ by the lemma. Conversely, by Proposition 4.2.1 and Theorem 4.2.2, $T = \sum_{j=0}^{J} c_j \delta_a^{(j)}$ for some J. By the lemma,

$$\langle (x-a)^n T,\, \phi \rangle = \sum_{j \geq n} (-1)^j \frac{j!}{(j-n)!}\, c_j\, \phi^{(j-n)}(a).$$

Since the functionals $\phi \mapsto \phi^{(i)}(a)$, $i = 0, 1, 2, \ldots$, are linearly independent, this vanishes for all ϕ only if $c_j = 0$ for all $j \geq n$. □

For example, the solutions of $x^2 T = 0$ are exactly $T = a\delta_0 + b\delta_0'$.

> *Remark 4.3.3.* Generalizing to several variables is awkward. The problem $|x|^n T = 0$ is ill-posed for odd n, since $|x|^n$ is not smooth. One might instead consider smooth functions vanishing to high order at the origin, such as xy on $\mathbb{R}^2$; but xy vanishes on both coordinate axes, so solutions of $xyT = 0$ can be supported on a large set and are much wilder.

We say a smooth $f : \mathbb{R} \to \mathbb{C}$ has a zero of ***order*** $k < \infty$ at z if $f(z) = \cdots = f^{(k-1)}(z) = 0 \neq f^{(k)}(z)$; equivalently, $\lim_{x \to z} f(x)/(x-z)^k$ exists and is nonzero.

Lemma 4.3.4
If every zero of a smooth $f : \mathbb{R} \to \mathbb{C}$ has finite order, then the zero set Z of f has no accumulation points. In particular Z is closed, discrete and at most countable.

Proof. Suppose zeros $z_n \to z$ with $z_n \neq z$. Let u be either $\Re f$ or $\Im f$; then u is a real smooth function vanishing at every z_n. By continuity $u(z) = 0$. Between consecutive z_n, Rolle's theorem yields zeros of u', which also accumulate at z; hence $u'(z) = 0$. Repeating, $u^{(k)}(z) = 0$ for every k. Applying this to both $\Re f$ and $\Im f$ shows that $f^{(k)}(z) = 0$ for every k, so z is a zero of infinite order, a contradiction. □

Theorem 4.3.5 (Division by a function with zeros of finite order)
Let $f : \mathbb{R} \to \mathbb{C}$ be smooth, and suppose its zero set Z consists of zeros of finite order, $z \in Z$ having order k_z. Then $fT = 0$ iff

$$T = \sum_{z \in Z} D_z \delta_z,$$

where each D_z is a constant-coefficient differential operator of order at most $k_z - 1$. (The sum is locally finite by Lemma 4.3.4, and Z may be finite or countably infinite.)

Proof. By Lemma 4.3.4 we may choose functions $\chi_z \in C_c^\infty$, $z \in Z$, with pairwise disjoint supports, each $\chi_z \equiv 1$ near z, and with the family locally finite. By Proposition 4.2.1, $\operatorname{supp} T \subseteq Z$. Since $1 - \sum_z \chi_z$ vanishes on a neighbourhood of Z, we get $T = \sum_z \chi_z T$, a locally finite sum. Each $\chi_z T$ is supported in $\{z\}$, so $\chi_z T = \sum_{j \leq J} c_j \delta_z^{(j)}$ by Theorem 4.2.2.

Near z write $f(x) = (x-z)^{k_z} g(x)$ with g smooth and $g(z) \neq 0$ (Taylor's theorem with integral remainder). Then $0 = \chi_z f T = f\,(\chi_z T)$. Choose $\eta \in C_c^\infty$ supported where $g \neq 0$, with $\eta \equiv 1$ near z; multiplying by the smooth function η/g gives $(x-z)^{k_z} \eta\, \chi_z T = (x-z)^{k_z} \chi_z T = 0$ (multiplying a distribution supported at z by a function equal to 1 near z changes nothing). By Proposition 4.3.2, $\chi_z T$ is a combination of $\delta_z, \ldots, \delta_z^{(k_z-1)}$. The converse direction follows from Lemma 4.3.1 and the same factorization. □

Example 4.3.6
For $f(x) = (x-a)(x-b)$ with $a \neq b$, both zeros are simple, so the solutions of $fT = 0$ are exactly $T = A\delta_a + B\delta_b$. For $f(x) = \sin(\pi x)$, all zeros are simple and located at the integers, so the solutions of $\sin(\pi x)\, T = 0$ are $T = \sum_{n\in\mathbb{Z}} c_n \delta_n$ for arbitrary constants c_n. This last case will give us the Poisson summation formula (Section 11.1).

Going further than this leads to genuinely difficult territory, and we stop here.

5 The Fourier Transform

We finally come to the reason for introducing tempered distributions.

5.1 The Fourier transform on functions

Fourier transform

The Fourier transform is the linear map $\mathcal{F} : \mathcal{S}(\mathbb{R}^N) \to \mathcal{S}(\mathbb{R}^N)$,

$$\hat{u}(\xi) = (\mathcal{F}u)(\xi) := \int_{\mathbb{R}^N} u(x)\, e^{-2\pi i \xi \cdot x}\, \mathrm{d}x, \tag{5.1}$$

and the inverse Fourier transform is $\check{u}(x) = \mathcal{F}^{-1}u(x) := \int_{\mathbb{R}^N} u(\xi)\, e^{2\pi i x \cdot \xi}\, \mathrm{d}\xi$.

That $\mathcal{F}$ maps $\mathcal{S}$ to $\mathcal{S}$ is well documented, and we collect the key facts without proof.

- $\mathcal{F}$ is continuous on $\mathcal{S}$: if $f_k \to f$ in $\mathcal{S}$ (i.e. $\|f_k - f\|_{m,n} \to 0$ for all m, n) then $\hat{f}_k \to \hat{f}$ in $\mathcal{S}$. $\mathcal{F}^{-1}$ is its inverse on $\mathcal{S}$, with the same properties.
- Formula (5.1) makes sense for $u \in L^1$ and gives $\mathcal{F} : L^1 \to L^\infty$. By the ***Riemann–Lebesgue lemma***, in fact $\hat{u}$ is continuous and tends to 0 at infinity.
- On L^2, formula (5.1) is meaningless in general (the integral need not converge), but $\mathcal{F}$ extends from $\mathcal{S}$ by density to a unitary map $L^2 \to L^2$ (***Plancherel***). Interpolating, $\mathcal{F} : L^p \to L^{p'}$ for $1 \le p \le 2$ (***Hausdorff–Young***), where p' is the Hölder conjugate.
- For $p > 2$ the Fourier transform of an L^p function is in general not a function at all. Tempered distributions will rescue us here.
- For $u \in L^1$ one cannot naively write $u(x) = \int \hat{u}(\xi) e^{2\pi i x \cdot \xi}\, \mathrm{d}\xi$, because $\hat{u}$ need not be integrable. The inversion formula holds when $\hat{u} \in L^1$ as well.
- $\overline{\hat{u}} = \check{\overline{u}}$ and $\overline{\check{u}} = \hat{\overline{u}}$ ("the arrow flips as it passes under the bar"). This is why $\mathcal{F}$ and $\mathcal{F}^{-1}$ have the same qualitative properties.
- The Gaussian $e^{-\pi|x|^2}$ is its own Fourier transform.
- (Uncertainty) If $u \neq 0$, then u and $\hat{u}$ cannot both be compactly supported.
- (Decay $\leftrightarrow$ smoothness) If u is compactly supported, $\hat{u}$ is smooth. The converse fails.
- (Derivatives become multiplication) For $u \in \mathcal{S}$, $\widehat{\partial_{x_j} u}(\xi) = 2\pi i \xi_j\, \hat{u}(\xi)$, and more generally $\widehat{\partial^\alpha u}(\xi) = (2\pi i \xi)^\alpha \hat{u}(\xi)$. This is why the Fourier transform can "destroy" constant-coefficient PDEs.
- (Convolution becomes multiplication) $\widehat{u * v} = \hat{u}\, \hat{v}$.
- (Translation becomes modulation) If $g(x) = f(x - a)$ then $\hat{g}(\xi) = e^{-2\pi i \xi \cdot a} \hat{f}(\xi)$.

- (Multiplication formula) For $u, v \in \mathcal{S}$ (or L^2),

$$\int_{\mathbb{R}^N} \hat{u}\, v \, \mathrm{d}x = \int_{\mathbb{R}^N} u\, \hat{v} \, \mathrm{d}x, \qquad \int \hat{u}\, \check{v} = \int uv, \qquad \int \hat{u}\, \overline{\hat{v}} = \int u\, \bar{v}.$$

  With $(u, v) := \int u\bar{v}$ the $L^2(\mathbb{R}^N; \mathbb{C})$ inner product, the last identity reads $(u, v) = (\hat{u}, \hat{v})$: $\mathcal{F}$ is an isometry of L^2. Taking $u = v$ gives ***Parseval's identity*** $\int |u|^2 = \int |\hat{u}|^2$.

5.2 The Fourier transform of a tempered distribution

Requiring integrability or smoothness of u before we may take $\hat{u}$ is restrictive; it forbids transforming many functions to which we would like to apply Fourier analysis. Tempered distributions remove this restriction.

The multiplication formula $\int \hat{u}\, v = \int u\, \hat{v}$ says that $\langle \hat{u}, v \rangle = \langle u, \hat{v} \rangle$ for functions. We adopt this as a definition.

Fourier transform of a tempered distribution

For $T \in \mathcal{S}'(\mathbb{R}^N)$, the tempered distribution $\hat{T}$ is defined by

$$\langle \hat{T}, \phi \rangle := \langle T, \hat{\phi} \rangle, \qquad \phi \in \mathcal{S}(\mathbb{R}^N).$$

Similarly $\langle \check{T}, \phi \rangle := \langle T, \check{\phi} \rangle$.

In short: move the hat across the bracket.

- $\hat{T}$ is indeed tempered, because $\mathcal{F}$ is continuous on $\mathcal{S}$.
- The definition makes no sense for general distributions: if $\phi \in C_c^\infty$, then $\hat{\phi}$ is never in C_c^∞ (unless $\phi = 0$), so $\langle T, \hat{\phi} \rangle$ is undefined. This is exactly why the Schwartz class, on which $\mathcal{F}$ is a bijection, is the right space of test functions.
- When T is itself a Schwartz (or L^1) function, the definition agrees with the classical transform, by the multiplication formula.

Example 5.2.1

We have $\hat{\delta}_0 = 1$, since

$$\langle \hat{\delta}_0, \phi \rangle = \hat{\phi}(0) = \int_{\mathbb{R}^N} \phi(x) e^{-2\pi i 0 \cdot x} \, \mathrm{d}x = \int \phi = \langle 1, \phi \rangle.$$

Similarly $\hat{1} = \delta_0$, which is slightly harder from scratch (see Chapter 7). More generally

$$\hat{\delta}_a = e^{-2\pi i a \cdot \xi}, \qquad \widehat{e^{2\pi i a \cdot x}} = \delta_a,$$

which also follows from the translation rule below.

5.3 Properties

Most algebraic properties of the classical transform carry over.

- $T \mapsto \hat{T}$ is linear.

- $\widehat{\partial^\alpha T} = (2\pi i\xi)^\alpha \hat{T}$.
- $\check{\hat{T}} = \hat{\check{T}} = T$.
- Let $\tau_a T$ denote the translate "$T(x-a)$", defined by $\langle \tau_a T, \phi\rangle := \langle T, \phi(\cdot + a)\rangle$. Then $\widehat{\tau_a T} = e^{-2\pi i\xi\cdot a}\,\hat{T}$.

Other properties, such as Parseval's identity, have no good analogue. One remarkable property survives, however.

Theorem 5.3.1 (Paley–Wiener–Schwartz)
If T is a distribution with compact support, then T is tempered and $\hat{T}$ is a smooth function, given by $\hat{T}(\xi) = \langle T, e^{-2\pi ix\cdot\xi}\rangle$. In fact $\hat{T}$ extends to an entire function on $\mathbb{C}^N$.

- This vastly generalizes the fact that compactly supported *functions* have smooth Fourier transforms. For functions one simply differentiates under the integral; for distributions the proof is harder.
- Corollary: if $\hat{T}$ has compact support, then T is a smooth function.
- Example: δ_0 has compact support, and indeed $\hat{\delta}_0 = 1$ is smooth.

5.4 Fourier transforms of measures

A tempered measure μ has a Fourier transform: $\langle \hat{\mu}, \phi\rangle = \langle \mu, \hat{\phi}\rangle = \int \hat{\phi}\,d\mu$. If μ is *finite*, Fubini's theorem lets us go further:

$$\langle \hat{\mu}, \phi\rangle = \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} e^{-2\pi ix\cdot\xi}\phi(\xi)\,d\xi\,d\mu(x) = \int_{\mathbb{R}^N}\Big(\int_{\mathbb{R}^N} e^{-2\pi ix\cdot\xi}\,d\mu(x)\Big)\phi(\xi)\,d\xi.$$

Proposition 5.4.1
If μ is a finite measure, then $\hat{\mu}$ is the bounded continuous function

$$\hat{\mu}(\xi) = \int_{\mathbb{R}^N} e^{-2\pi ix\cdot\xi}\,d\mu(x).$$

Proof. $|\hat{\mu}(\xi)| \le |\mu|(\mathbb{R}^N)$. If $\xi_n \to \xi$, the integrands converge pointwise and are bounded by 1, which is $|\mu|$-integrable; dominated convergence gives $\hat{\mu}(\xi_n) \to \hat{\mu}(\xi)$. □

This generalizes the Riemann–Lebesgue lemma, except that decay at infinity is lost: $\hat{\delta}_0 = 1$ is continuous but does not tend to 0.

5.5 Examples

Example 5.5.1 (Cosine and sine)
Since $\cos(2\pi x) = \frac{1}{2}(e^{2\pi ix} + e^{-2\pi ix})$,

$$\widehat{\cos(2\pi x)} = \tfrac{1}{2}\delta_1 + \tfrac{1}{2}\delta_{-1}, \qquad \widehat{\sin(2\pi x)} = \tfrac{1}{2i}\delta_1 - \tfrac{1}{2i}\delta_{-1}.$$

Example 5.5.2 (Surface measure on the sphere)
Let $\mathrm{d}\omega = \mathcal{H}^{N-1} \llcorner \partial B(0,1)$ be surface measure on the unit sphere. It is compactly supported, so by Theorem 5.3.1 $\widehat{\mathrm{d}\omega}$ is the smooth function

$$\widehat{\mathrm{d}\omega}(\xi) = \int_{\partial B(0,1)} e^{-2\pi i \xi \cdot x} \, \mathrm{d}\mathcal{H}^{N-1}(x).$$

The *method of stationary phase* yields the decay estimate $|\widehat{\mathrm{d}\omega}(\xi)| \leq C|\xi|^{-(N-1)/2}$, which is fundamental in harmonic analysis.

Example 5.5.3 (The arctangent)
$\arctan x$ is not integrable, so the integral formula cannot be used. A tempting symbolic computation runs

$$\widehat{\arctan}(\xi) = \frac{1}{2\pi i \xi}\Big(\frac{\mathrm{d}}{\mathrm{d}x} \arctan x\Big)^{\wedge} = \frac{1}{2\pi i \xi}\Big(\frac{1}{1+x^2}\Big)^{\wedge}.$$

Now $(1+x^2)^{-1}$ is integrable, and a residue computation gives

$$\Big(\frac{1}{1+x^2}\Big)^{\wedge}(\xi) = \pi e^{-2\pi|\xi|},$$

suggesting $\widehat{\arctan}(\xi) = \frac{1}{2i} e^{-2\pi|\xi|}/\xi$. This is *almost* right. To make it rigorous, regard arctan as a tempered distribution (it is bounded). Then

$$2\pi i \xi \cdot \widehat{\arctan} = \Big(\frac{\mathrm{d}}{\mathrm{d}x} \arctan x\Big)^{\wedge} = \pi e^{-2\pi|\xi|}$$

holds rigorously. Multiplying by $1/(2\pi i \xi)$ is illegal (it is not smooth). Instead, observe that $\pi e^{-2\pi|\xi|} = 2\pi i \xi \cdot \frac{1}{2i}\,\mathrm{p.v.}\,\frac{e^{-2\pi|\xi|}}{\xi}$, so

$$\xi\Big(\widehat{\arctan} - \frac{1}{2i}\,\mathrm{p.v.}\,\frac{e^{-2\pi|\xi|}}{\xi}\Big) = 0,$$

and Proposition 4.1.1 gives $\widehat{\arctan} - \frac{1}{2i}\,\mathrm{p.v.}\,\frac{e^{-2\pi|\xi|}}{\xi} = a\delta_0$ for a constant a. To find a, test against the self-dual Gaussian $g(\xi) = e^{-\pi\xi^2}$: $\Big\langle \widehat{\arctan}, g \Big\rangle = \langle \arctan, \hat{g} \rangle = \int \arctan(x) e^{-\pi x^2}\,\mathrm{d}x = 0$ by oddness, and the principal value term also vanishes by oddness. Hence $a = 0$ and

$$\boxed{\widehat{\arctan}(\xi) = \frac{1}{2i}\,\mathrm{p.v.}\,\frac{e^{-2\pi|\xi|}}{\xi}}$$

6 Convergence and Convolution

6.1 Convergence in distribution

Convergence in distribution
Distributions T_n ***converge in distribution*** to T if $\langle T_n, \phi\rangle \to \langle T, \phi\rangle$ for every test function ϕ. In functional analytic terms this is weak-$*$ convergence, $T_n \overset{*}{\rightharpoonup} T$.

This notion is extremely weak, as the following comparisons show.

6.1.1 Convergence of functions

- If $f_n \to f$ in L^p_{loc} for some $1 \le p \le \infty$, then $f_n \to f$ in distribution.
- If $f_n \rightharpoonup f$ weakly in L^p $(1 \le p < \infty)$, or $f_n \overset{*}{\rightharpoonup} f$ in L^p $(1 < p \le \infty)$, then $f_n \to f$ in distribution.
- Uniform convergence implies convergence in distribution.
- Almost-everywhere convergence does *not* (Section 12.2, Problem P4).

6.1.2 Convergence of measures

For Radon measures, ***weak-$*$ convergence*** $\mu_n \overset{*}{\rightharpoonup} \mu$ means $\int f\,\mathrm{d}\mu_n \to \int f\,\mathrm{d}\mu$ for every continuous compactly supported f. Since smooth test functions are in particular continuous, weak-$*$ convergence implies convergence in distribution. For *positive* measures the converse holds too.

Proposition 6.1.1
Let μ_n, μ be positive Radon measures (or signed ones with $\sup_n |\mu_n|(K) < \infty$ for every compact K). Then $\mu_n \to \mu$ in distribution iff $\mu_n \overset{*}{\rightharpoonup} \mu$.

Proof. One direction is immediate, so assume $\mu_n \to \mu$ in distribution and fix $f \in C_c$. Since smooth functions are dense in the uniform norm, there is $\phi \in C_c^\infty$ with $\|f - \phi\|_\infty < \varepsilon$; enlarging supports slightly, choose a single compact K containing the supports of both f and ϕ. Then

$$\begin{aligned}\left|\int_{\mathbb{R}^N} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}^N} f\,\mathrm{d}\mu\right| &\le \int_K |f-\phi|\,\mathrm{d}|\mu_n| + \left|\int_K \phi\,\mathrm{d}\mu_n - \int_K \phi\,\mathrm{d}\mu\right| + \int_K |\phi - f|\,\mathrm{d}|\mu| \\ &\le \varepsilon\,|\mu_n|(K) + \left|\,\langle \mu_n - \mu, \phi\rangle\,\right| + \varepsilon\,|\mu|(K).\end{aligned}$$

The middle term tends to 0 by hypothesis. The local masses $|\mu_n|(K)$ are bounded by some C — by assumption in the signed case, and automatically in the positive case: choosing $\chi \in C_c^\infty$ with $0 \le \chi \le 1$ and $\chi \equiv 1$ on K gives $\mu_n(K) \le \langle \mu_n, \chi\rangle \to \langle \mu, \chi\rangle$. Hence

$$\limsup_{n\to\infty} \left|\int f\,\mathrm{d}\mu_n - \int f\,\mathrm{d}\mu\right| \le \big(C + |\mu|(K)\big)\,\varepsilon,$$

and $\varepsilon > 0$ was arbitrary. □

Remark 6.1.2. The local bound on $|\mu_n|(K)$ is what makes the approximation step work, so it is worth stating explicitly. For positive measures it is free, as the proof shows; for signed measures the total variation can in principle blow up on a compact set while the distributional limit still exists.

6.1.3 Convergence in probability theory

Recall that $X_n \to X$ in distribution if $\mathbb{E}f(X_n) \to \mathbb{E}f(X)$ for every bounded continuous f, i.e.

$$\int_{\mathbb{R}} f \, d\mu_{X_n} \to \int_{\mathbb{R}} f \, d\mu_X \qquad \text{for all } f \in C_b(\mathbb{R}),$$

where $\mu_X = \mathbb{P} \circ X^{-1}$ is the law of X. The class C_b is much larger than C_c, let alone C_c^∞. Nevertheless it may be replaced by C_c^∞.

Proposition 6.1.3
Let μ_n, μ be probability measures on $\mathbb{R}$ with $\int f \, d\mu_n \to \int f \, d\mu$ for every $f \in C_c(\mathbb{R})$. Then the same holds for every $f \in C_b(\mathbb{R})$.

Proof. Fix $f \in C_b$ and $\varepsilon > 0$. Since μ is a probability measure there is $M > 0$ with $\mu([-M, M]) \geq 1 - \varepsilon$. Choose $\phi \in C_c$ with $\phi = f$ on $[-M-1, M+1]$, $\|\phi\|_\infty \leq \|f\|_\infty$ and $\operatorname{supp}\phi \subseteq [-M-1, M+1]$. Then

$$\left| \int f \, d\mu_n - \int f \, d\mu \right| \leq \underbrace{\int |f - \phi| \, d\mu_n}_{I} + \underbrace{\left| \int \phi \, d\mu_n - \int \phi \, d\mu \right|}_{II} + \underbrace{\int |\phi - f| \, d\mu}_{III}.$$

Both $f - \phi$ and $\phi - f$ vanish on $[-M-1, M+1]$, so

$$I \leq \|f\|_\infty \big(1 - \mu_n([-M-1, M+1])\big),$$
$$III \leq \|f\|_\infty \big(1 - \mu([-M-1, M+1])\big) \leq \|f\|_\infty \varepsilon,$$

while $II \to 0$ by hypothesis. Hence

$$\limsup_{n\to\infty} \left| \int f \, d\mu_n - \int f \, d\mu \right| \leq \|f\|_\infty \varepsilon + \|f\|_\infty \Big(1 - \liminf_{n\to\infty} \mu_n([-M-1, M+1])\Big).$$

It remains to show $\liminf_n \mu_n([-M-1, M+1]) \geq \mu([-M, M])$, for then the right side is at most $2\|f\|_\infty \varepsilon$ and we are done. Take a smooth ψ equal to 1 on $[-M, M]$, equal to 0 outside $[-M-1, M+1]$, and with values in $[0, 1]$ in between. Then $\psi \in C_c$, so

$$\liminf_{n\to\infty} \mu_n([-M-1, M+1]) \geq \lim_{n\to\infty} \int \psi \, d\mu_n = \int \psi \, d\mu \geq \mu([-M, M]). \qquad \square$$

Combining this with Proposition 6.1.1, the two uses of the phrase agree: $X_n \to X$ in distribution if and only if the laws μ_{X_n} converge to μ_X in distribution.

6.2 Convolving a distribution with a function

For functions, $(f * g)(x) = \int f(y) g(x - y) \, dy$. Testing against ϕ and rearranging so that f appears alone on one side of the bracket,

$$\langle f * g, \phi \rangle = \iint f(y) g(x - y) \phi(x) \, dy \, dx = \Big\langle f, \; y \mapsto \int g(x - y) \phi(x) \, dx \Big\rangle.$$

The inner expression is almost, but not quite, a convolution. Writing $\tilde{g}(x) := g(-x)$ for the reflection of g, it equals $\int \tilde{g}(y - x) \phi(x) \, dx = (\tilde{g} * \phi)(y)$. So $\langle f * g, \phi \rangle = \langle f, \tilde{g} * \phi \rangle$, which suggests the general definition.

Distribution $*$ test function
If T is a distribution and $g \in C_c^\infty$, or T is tempered and $g \in \mathcal{S}$,

$$\langle T * g, \phi \rangle := \langle T, \tilde{g} * \phi \rangle .$$

One can check that $T * g$ is a (tempered) distribution, but more is true.

Proposition 6.2.1
The convolution of a distribution with a C_c^∞ function, or of a tempered distribution with a Schwartz function, is a C^∞ function, namely $(T * g)(x) = \langle T, g(x - \cdot) \rangle$.

The result need not have compact support: with $T = 1$ and $g \in C_c^\infty$, $T * g$ is the constant $\int g$.

6.3 Convolving two distributions

In general two distributions cannot be convolved. The obstruction is integrability: $1 * 1$ is meaningless. But if one of them has compact support, we can define the convolution using associativity.

Distribution $*$ compactly supported distribution
Let S, T be distributions (or tempered distributions) with T compactly supported. Then $S * T$ is the unique distribution (tempered distribution) with

$$(S * T) * \phi = S * (T * \phi) \qquad \text{for all } \phi \in C_c^\infty \text{ (resp. } \mathcal{S}).$$

This works because $T * \phi$ is smooth and, T being compactly supported, also compactly supported. The definition also works if S is the compactly supported one (Problem P6), so $S * T = T * S$ is meaningful as soon as either factor has compact support.

6.3.1 A heuristic for when convolution makes sense

Heuristically, $S * T$ makes sense iff the pairing $\langle S, T * \phi \rangle$ makes sense. This suggests a three-step protocol.

1. Determine how badly behaved $T * \phi$ can be as ϕ ranges over test functions: is it in C_c^∞ (best), $\mathcal{S}$, C_0^∞ (smooth and vanishing at infinity), or merely C^∞ (no information)?
2. Find the largest natural class of functions ψ to which the pairing $\langle S, \psi \rangle$ extends. If $T * \phi$ always lies in that class, proceed.
3. Then $S * T$ is sensible.

Example 6.3.1
Let μ be a finite measure. For $\phi \in C_c^\infty$, $\mu * \phi$ is generally not in C_c^∞, but it lies in C_0^∞ (smooth, with all derivatives vanishing at infinity, by dominated convergence). So $S * \mu$ makes sense whenever $\langle S, \psi \rangle$ makes sense for all $\psi \in C_0^\infty$. If $S = \nu$ is another finite measure, $\int \psi \, \mathrm{d}\nu$ converges since $\int |\psi| \, \mathrm{d}|\nu| \leq \|\psi\|_\infty |\nu|(\mathbb{R}^N)$ — and here both $\psi \in C_0$ and the finiteness of ν were essential. So $\nu * \mu$ makes sense when both are finite; the heuristic correctly predicts that it can fail when only one is.

6.3.2 Regularity versus structure

We generally expect $S * T$ to be at least as regular as the more regular of S and T: a continuous function convolved with a measure, one of them compactly supported, should be continuous. But *regularity* (smoothness) and *structure* (being a measure or a function) are different things. It is tempting to believe that a function convolved with a distribution is a function. This is false.

Example 6.3.2
On $\mathbb{R}$, take $f = \mathbf{1}_{[0,1]}$ and $T = \delta_0'$, both compactly supported. Then $(\tilde{f} * \phi)(x) = \int \mathbf{1}_{[0,1]}(y - x)\phi(y)\,\mathrm{d}y = \int_x^{x+1} \phi$, and

$$\left\langle \mathbf{1}_{[0,1]} * \delta_0',\, \phi \right\rangle = \langle \delta_0',\, \tilde{f} * \phi \rangle = -\frac{\mathrm{d}}{\mathrm{d}x}\Big|_{x=0} \int_x^{x+1} \phi(y)\,\mathrm{d}y = \phi(0) - \phi(1).$$

So $\mathbf{1}_{[0,1]} * \delta_0' = \delta_0 - \delta_1$, which is not a function.

6.3.3 Function $*$ measure

Let $f \in L^1_{\mathrm{loc}}$ and μ Radon. Then $f * \mu$ makes sense under any of the following (non-exhaustive) hypotheses:

- μ is compactly supported ($\mu * C_c^\infty \subseteq C_c^\infty$, on which f acts);
- f is compactly supported ($\mu * C_c^\infty \subseteq C^\infty$, on which f acts);
- $f \in \mathcal{S}$ and μ is tempered ($f * \mathcal{S} \subseteq \mathcal{S}$, on which μ acts);
- $f \in L^1$ and μ is finite ($\mu * C_c^\infty \subseteq C_0^\infty$, on which f acts).

In each case

$$\langle f * \mu,\, \phi \rangle = \langle \mu,\, \tilde{f} * \phi \rangle = \iint f(y - x)\phi(y)\,\mathrm{d}y\,\mathrm{d}\mu(x) = \int \Big(\int f(y - x)\,\mathrm{d}\mu(x) \Big) \phi(y)\,\mathrm{d}y,$$

so $f * \mu$ is the *function* $(f * \mu)(y) = \int_{\mathbb{R}^N} f(y - x)\,\mathrm{d}\mu(x)$.

6.3.4 Function $*$ distribution

Let $f \in L^1_{\mathrm{loc}}$ and let T be a distribution. We already know from the first definition that $f * T$ is sensible for $f \in C_c^\infty$, and is in fact smooth. Which smooth function is it? Pretend for a moment that T is a function and compute formally:

$$\begin{aligned}
\langle f * T,\, \phi \rangle &= \langle T,\, \tilde{f} * \phi \rangle = \left\langle T,\, \int_y f(y - x)\phi(y)\,\mathrm{d}y \right\rangle \\
&= \int_x T(x) \int_y f(y - x)\phi(y)\,\mathrm{d}y\,\mathrm{d}x = \int_y \Big(\int_x T(x) f(y - x)\,\mathrm{d}x \Big) \phi(y)\,\mathrm{d}y \\
&= \int_y \langle T,\, f(\,\cdot - x)\rangle\, \phi(y)\,\mathrm{d}y = \Big\langle \langle T,\, f(\,\cdot - x)\rangle\,,\, \phi(y) \Big\rangle.
\end{aligned}$$

So we expect $f * T$ to be the function $x \mapsto \langle T,\, f(\,\cdot - x)\rangle$. Making this rigorous is not obvious, because the only illegitimate step above — moving the integral outside the bracket — is exactly the point.

Proof sketch. The step to justify is

$$\left\langle T, \int_y f(y-x)\phi(y)\,\mathrm{d}y \right\rangle = \int_y \langle T,\, f(y-x)\phi(y)\rangle\,\mathrm{d}y.$$

The trick is that the integral may be approximated by Riemann sums, over which a distribution passes by linearity. Roughly,

$$\begin{aligned}\langle f * T,\, \phi\rangle = \langle T,\, \tilde{f} * \phi\rangle &= \left\langle T, \int_{\mathbb{R}^N} f(y-x)\phi(y)\,\mathrm{d}y \right\rangle \\ &\approx \left\langle T,\, h^N \sum_{y\in h\mathbb{Z}^N} f(y-x)\phi(y) \right\rangle = h^N \sum_{y\in h\mathbb{Z}^N} \langle T,\, f(y-x)\rangle\,\phi(y) \\ &\approx \int_{\mathbb{R}^N} \langle T,\, f(y-x)\rangle\,\phi(y)\,\mathrm{d}y.\end{aligned}$$

Both approximations converge as $h \to 0$: the Riemann sums converge to the integral in the topology of C_c^∞, which is what lets T pass through the first one. □

If we only know that f or T is compactly supported, $f * T$ still makes sense, but it need not be a function, as the example above shows.

6.3.5 Measure $*$ measure

The convolution of *finite* Radon measures makes sense, a larger class than the compactly supported ones; and if only one of them is finite, $\mu * \nu$ can be meaningless. For functions,

$$\begin{aligned}\langle f * g,\, \phi\rangle &= \iint f(x-y)g(y)\phi(x)\,\mathrm{d}y\,\mathrm{d}x = \iint \phi(x+y)f(x)g(y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \iint \phi(x+y)\,\mathrm{d}\mu_f(x)\,\mathrm{d}\mu_g(y),\end{aligned}$$

and correspondingly, for finite measures,

$$\langle \mu * \nu,\, \phi\rangle = \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} \phi(x+y)\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y).$$

As observed in Example 2.3.6, the right side involves no derivatives of ϕ, so $\mu * \nu$ has order 0 and is a measure. Taking $\phi = \mathbf{1}_E$,

$$(\mu * \nu)(E) = \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} \mathbf{1}_E(x+y)\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y), \tag{6.1}$$

and $|\mu * \nu|(\mathbb{R}^N) \le |\mu|(\mathbb{R}^N)\,|\nu|(\mathbb{R}^N)$, so $\mu * \nu$ is again finite. Two measures may even convolve to a function; see Example 7.4.1.

6.3.6 Measure $*$ distribution

Since a function convolved with a distribution is in general no better than a distribution, there seems to be nothing special to say here.

6.4 General properties

Commutativity and associativity. By Rudin [3, Thm. 6.37]:

- if S or T has compact support, then $S * T = T * S$;
- if at least two of S, T, R have compact support, then $(S * T) * R = S * (T * R)$.

⚠ It is not enough that both sides are defined. With $H = \mathbf{1}_{(0,\infty)}$,

$$1 * (\delta_0' * H) = 1 * \delta_0 = 1, \qquad \text{but} \qquad (1 * \delta_0') * H = 0 * H = 0.$$

Fourier transform. If S, T are tempered and T has compact support, then $\widehat{S * T} = \hat{S}\,\hat{T}$. The right side looks like a forbidden product of distributions, but $\hat{T}$ is a smooth function by Theorem 5.3.1.

Derivatives. For $f, g \in C_c^\infty$, $\partial_{x_i}(f * g) = (\partial_{x_i} f) * g = f * (\partial_{x_i} g)$. Likewise, if S, T are distributions and one has compact support,

$$\partial^\alpha (S * T) = (\partial^\alpha S) * T = S * (\partial^\alpha T).$$

6.5 Mollification

Let $\varphi \in C_c^\infty$ with $\varphi \geq 0$ and $\int \varphi = 1$, and set $\varphi_\varepsilon(x) := \varepsilon^{-N}\varphi(x/\varepsilon)$. (The precise choice does not matter.) As $\varepsilon \to 0$, φ_ε concentrates at the origin, and indeed $\varphi_\varepsilon \to \delta_0$ in distribution; for this reason (φ_ε) is called an ***approximation to the identity***. It follows that

$$T * \varphi_\varepsilon \longrightarrow T * \delta_0 = T \quad \text{in distribution}$$

for every distribution T. The smooth function $T * \varphi_\varepsilon$ is the ***mollification*** of T.

- If $T = f$ is a function, this recovers weak convergence of mollifiers, which is not news: $f * \varphi_\varepsilon \to f$ in much stronger senses, e.g. in L^1_{loc}.
- If $T = \mu$ is a measure, we recover the more interesting fact that the regularizations $\mu * \varphi_\varepsilon$ converge weakly-$*$ to μ.

Mollification allows us, in many arguments, to treat distributions as if they were smooth and pass to the limit. Put differently, smooth functions are dense in the distributions. From the chain of inclusions

$$C_c^\infty(\mathbb{R}^N) \subseteq \mathcal{S}(\mathbb{R}^N) \subseteq \mathcal{S}'(\mathbb{R}^N) \subseteq \mathcal{D}'(\mathbb{R}^N)$$

one expects (and, after checking that the topologies are compatible, one can prove) that each inclusion is dense: Schwartz functions are dense in tempered distributions, and tempered distributions are dense in distributions.

7 More Examples

7.1 Fourier transforms of deltas and exponentials

Fix $a \in \mathbb{R}^N$. Since δ_a has compact support, Theorem 5.3.1 says its transform is the function

$$\hat{\delta}_a(\xi) = \left\langle \delta_a,\, e^{-2\pi i x\cdot\xi} \right\rangle = e^{-2\pi i a\cdot\xi}.$$

What is the transform of $e^{-2\pi i a\cdot x}$? It is a bounded function, hence tempered, but it lies in neither L^1 nor L^2, so we should not expect a function. Testing directly against $\phi \in \mathcal{S}$,

$$\left\langle \widehat{e^{-2\pi i a\cdot x}},\, \phi \right\rangle = \int_{\mathbb{R}^N} e^{-2\pi i a\cdot x}\hat{\phi}(x)\,dx = \mathcal{F}^{-1}(\hat{\phi})(-a) = \phi(-a) = \langle \delta_{-a},\, \phi\rangle\,,$$

so $\widehat{e^{-2\pi i a\cdot x}} = \delta_{-a}$. We obtain the cycle

$$\hat{\delta}_a = e^{-2\pi i a\cdot\xi}, \qquad \widehat{e^{-2\pi i a\cdot x}} = \delta_{-a}, \qquad \hat{\delta}_{-a} = e^{2\pi i a\cdot\xi}, \qquad \widehat{e^{2\pi i a\cdot x}} = \delta_a.$$

There is a slicker route, via the following identity.

Proposition 7.1.1
For $\phi \in \mathcal{S}(\mathbb{R}^N)$, $\hat{\hat{\phi}}(x) = \phi(-x)$. Consequently $\hat{\hat{T}} = \tilde{T}$ for every tempered distribution T, where $\left\langle \tilde{T},\, \phi \right\rangle := \langle T,\, \phi(-\cdot)\rangle$.

Proof. $\hat{\hat{\phi}}(x) = \int \hat{\phi}(\xi)e^{-2\pi i x\cdot\xi}\,d\xi = \mathcal{F}^{-1}\hat{\phi}(-x) = \phi(-x)$. For distributions, $\left\langle \hat{\hat{T}},\, \phi \right\rangle = \left\langle T,\, \hat{\hat{\phi}} \right\rangle = \langle T,\, \phi(-\cdot)\rangle = \left\langle \tilde{T},\, \phi \right\rangle$. □

Hence $\widehat{e^{-2\pi i a\cdot x}} = \hat{\hat{\delta}}_a = \tilde{\delta}_a = \delta_{-a}$, with no computation at all.

Both routes above lean on Fourier inversion. It is instructive to see the same result obtained from first principles, since the computation shows exactly where the difficulty lies.

Computing $\widehat{e^{-2\pi i a\cdot x}}$ by hand. Unwinding the definitions,

$$\begin{aligned}
\left\langle \widehat{e^{-2\pi i a\cdot x}}, \phi \right\rangle = \left\langle e^{-2\pi i a\cdot x}, \hat{\phi} \right\rangle &= \int_{\mathbb{R}^N} e^{-2\pi i a\cdot x}\hat{\phi}(x)\,dx \\
&= \int_{x\in\mathbb{R}^N}\int_{\xi\in\mathbb{R}^N} e^{-2\pi i a\cdot x}e^{-2\pi i\xi\cdot x}\phi(\xi)\,d\xi\,dx \\
&= \int_{x\in\mathbb{R}^N}\int_{\xi\in\mathbb{R}^N} e^{-2\pi i(\xi+a)\cdot x}\phi(\xi)\,d\xi\,dx \\
&= \int_{x\in\mathbb{R}^N}\int_{\xi\in\mathbb{R}^N} e^{-2\pi i\xi\cdot x}\phi(\xi-a)\,d\xi\,dx.
\end{aligned}$$

We would like to exchange the integrals, but the integrand is not integrable: there is no decay in x. So write the outer integral as a limit over cubes $Q(0,R) = [-R,R]^N$, where Fubini does apply:

$$= \lim_{R\to\infty}\int_{\xi\in\mathbb{R}^N}\phi(\xi-a)\int_{x\in Q(0,R)} e^{-2\pi i\xi\cdot x}\,dx\,d\xi$$

$$= \lim_{R\to\infty}\int_{\xi\in\mathbb{R}^N}\phi(\xi-a)\prod_{j=1}^{N}\int_{-R}^{R} e^{-2\pi i\xi_j x_j}\,dx_j\,d\xi = \lim_{R\to\infty}\int_{\xi\in\mathbb{R}^N}\phi(\xi-a)\prod_{j=1}^{N}\frac{\sin(2\pi\xi_j R)}{\pi\xi_j}\,d\xi.$$

Now

$$\prod_{j=1}^{N} \frac{\sin(2\pi\xi_j R)}{\pi\xi_j} = R^N \prod_{j=1}^{N} \frac{\sin(2\pi\xi_j R)}{\pi\xi_j R} =: R^N g(R\xi),$$

and $\int_{\mathbb{R}^N} g = \int_{\mathbb{R}^N} \prod_j \frac{\sin(2\pi\xi_j)}{\pi\xi_j}\,\mathrm{d}\xi = 1$ as an improper Riemann integral. So $R^N g(R\,\cdot)$ is an approximation to the identity, and

$$\int_{\xi\in\mathbb{R}^N} \phi(\xi - a) \prod_{j=1}^{N} \frac{\sin(2\pi\xi_j R)}{\pi\xi_j}\,\mathrm{d}\xi \longrightarrow \phi(0-a) = \phi(-a)$$

as $R \to +\infty$. Hence $\left\langle \widehat{e^{-2\pi i a\cdot x}},\, \phi \right\rangle = \phi(-a) = \langle \delta_{-a},\, \phi\rangle$. □

7.2 Fourier transforms of polynomials

The monomial x^n on $\mathbb{R}$ is tempered, so it has a Fourier transform. What is it? (Try it before reading on.)

Proposition 7.2.1
On $\mathbb{R}$, $\widehat{x^n} = \left(\frac{i}{2\pi}\right)^n \delta_0^{(n)}$. More generally, in $\mathbb{R}^N$, $\widehat{x^\alpha} = \left(\frac{i}{2\pi}\right)^{|\alpha|} \partial^\alpha \delta_0$.

Proof. For Schwartz u, differentiating under the integral gives $\partial_\xi^\alpha \hat{u} = \big((-2\pi i x)^\alpha u\big)^\wedge$, and by duality the same holds for tempered u. Apply this with $u = 1$, $\hat{1} = \delta_0$: $\partial^\alpha \delta_0 = (-2\pi i)^{|\alpha|}\widehat{x^\alpha}$, and $(-2\pi i)^{-1} = i/(2\pi)$. □

Here is a second proof in one dimension, which is a good advertisement for the division theory of Chapter 4.

Second proof. Since $\frac{\mathrm{d}^{n+1}}{\mathrm{d}x^{n+1}} x^n = 0$,

$$(2\pi i\xi)^{n+1}\,\widehat{x^n} = \left(\frac{\mathrm{d}^{n+1}}{\mathrm{d}x^{n+1}} x^n\right)^\wedge = \hat{0} = 0,$$

so $\xi^{n+1}\widehat{x^n} = 0$. By Proposition 4.3.2, $\widehat{x^n} = \sum_{k=0}^{n} c_k \delta_0^{(k)}$ for some constants c_k. To find them, take inverse Fourier transforms of each side, using $\big(\delta_0^{(k)}\big)^\vee = (-2\pi i x)^k$:

$$x^n = \sum_{k=0}^{n} c_k (-2\pi i x)^k.$$

Comparing coefficients, $c_k = 0$ for $0 \le k \le n-1$ and $c_n = (-2\pi i)^{-n}$, which is $(i/2\pi)^n$. Equivalently, in one line,

$$\widehat{x^n} = \frac{1}{(-2\pi i)^n}\big(\widehat{(-2\pi i x)^n}\cdot 1\big) = \frac{1}{(-2\pi i)^n}\frac{\mathrm{d}^n}{\mathrm{d}\xi^n}\hat{1} = \frac{1}{(-2\pi i)^n}\delta_0^{(n)}.$$ □

By linearity every polynomial has a Fourier transform supported at the origin. Conversely, by Theorem 4.2.2, any tempered distribution whose Fourier transform is supported at the origin is a polynomial. This observation is behind several of the applications (Liouville's theorem, polynomially bounded harmonic functions).

7.3 Convolution with deltas

We have $T * \delta_0 = T$: the delta is the identity for convolution. Moreover

$$T * \partial^\alpha \delta_0 = (\partial^\alpha T) * \delta_0 = \partial^\alpha T,$$

by moving the derivative across the convolution. For tempered T one can also see this on the Fourier side: $\widehat{T * \partial^\alpha \delta_0} = \hat{T}\,(2\pi i\xi)^\alpha \hat{\delta}_0 = \widehat{\partial^\alpha T}$. Convolving with δ_a translates: $T * \delta_a = \tau_a T$. In short, convolving T with an operation applied to δ_0 applies that operation to T.

7.4 Two segments convolve to a square

Example 7.4.1
In $\mathbb{R}^2$ let $\mu = \mathcal{H}^1\llcorner([0,1] \times \{0\})$ and $\nu = \mathcal{H}^1\llcorner(\{0\} \times [0,1])$ be length measure on a horizontal and a vertical unit segment. Both are finite, so by (6.1)

$$(\mu * \nu)(E) = \int_0^1 \int_0^1 \mathbf{1}_E\big((s,0) + (0,t)\big)\,\mathrm{d}s\,\mathrm{d}t = \int_{[0,1]^2} \mathbf{1}_E(s,t)\,\mathrm{d}(s,t) = \int_E \mathbf{1}_{[0,1]^2}.$$

Two measures, neither of which is a function, convolve to the function $\mathbf{1}_{[0,1]^2}$.

8 Distributions Varying in Time

Time-dependent PDEs call for one more tool. We often seek $u(x,t)$ with $x \in \mathbb{R}^N$ and t in an interval $I \subseteq \mathbb{R}$. Time can be treated as one more spatial variable, but it is often more flexible to *curry*: regard u as a map

$$I \to (\mathbb{R}^N \to \mathbb{R}), \qquad t \mapsto u(t) = u(\cdot, t),$$

a function of time whose values are functions of space. This captures the idea of "a function that evolves". It leads naturally to ***time-dependent distributions***, i.e. maps $t \mapsto T(t)$ from I to $\mathcal{D}'(\mathbb{R}^N)$. (One could instead model everything as distributions on $\mathbb{R}^N \times I$; the curried view is usually easier to think with.)

Continuity and differentiability in time
Let $T : I \to \mathcal{D}'(\mathbb{R}^N)$.

- $T(t) \to S$ as $t \to t_0$ if $\langle T(t), \phi \rangle \to \langle S, \phi \rangle$ for every test function ϕ. If $S = T(t_0)$, T is ***continuous in time*** at t_0.
- T is ***differentiable in time*** at t_0 if the difference quotients $\big(T(t) - T(t_0)\big)/(t - t_0)$ converge in distribution as $t \to t_0$; the limit is denoted $\partial_t T(t_0)$. If this holds for all t, then $t \mapsto \partial_t T(t)$ is again a time-dependent distribution.

Example 8.0.1 (A moving point mass)
Let $T(t) := \delta_t$ on $\mathbb{R}$. Then $\partial_t T(t) = -\delta_t'$. Indeed,

$$\left\langle \frac{\delta_{t+h} - \delta_t}{h}, \phi \right\rangle = \frac{\phi(t+h) - \phi(t)}{h} \longrightarrow \phi'(t) = -\langle \delta_t', \phi \rangle .$$

The minus sign reflects the sign in the definition of the distributional derivative.

8.1 Elementary properties

The following are immediate from the definitions.

- If T is differentiable in time, it is continuous in time; and the map equal to $\frac{T(t)-T(t_0)}{t-t_0}$ for $t \neq t_0$ and to $\partial_t T(t_0)$ at $t = t_0$ is continuous in time.
- T is continuous in time iff $t \mapsto \langle T(t), \phi \rangle$ is continuous for every ϕ.
- $\partial_t T(t_0)$ exists iff $\frac{\mathrm{d}}{\mathrm{d}t} \langle T(t), \phi \rangle$ exists at t_0 for every ϕ, and then $\left.\dfrac{\mathrm{d}}{\mathrm{d}t}\right|_{t=t_0} \langle T(t), \phi \rangle = \langle \partial_t T(t_0), \phi \rangle$.

The following lemma is less trivial and is what makes the theory work: a continuous family of distributions cannot behave too wildly.

Lemma 8.1.1 (Uniform boundedness)
Let I be a compact interval and $T : I \to \mathcal{D}'(\mathbb{R}^N)$ continuous in time. For each compact K there are j and C, *independent of* t, with

$$|\langle T(t), \phi\rangle| \le C \sum_{|\alpha|\le j} \|\partial^\alpha \phi\|_{C(K)} \quad \text{for all } t \in I \text{ and all } \phi \text{ supported in } K.$$

In particular the orders of the $T(t)$ over K are uniformly bounded.

Proof. The space $\mathcal{D}_K$ of smooth functions supported in K is a Fréchet space under the seminorms $\|\cdot\|_{C^j(K)}$. For each fixed ϕ, the continuous function $t \mapsto \langle T(t), \phi\rangle$ is bounded on the compact I, so the family $\{T(t)|_{\mathcal{D}_K}\}_{t\in I}$ is pointwise bounded. By the Banach–Steinhaus theorem for Fréchet spaces it is equicontinuous, which is exactly the claimed uniform estimate. □

The proof uses only pointwise boundedness, so continuity can be weakened accordingly; the same argument applies to convergent *sequences* of distributions.

8.2 Time-dependent test functions

Convergence of test functions
A sequence $\phi_n \in C_c^\infty$ converges to ϕ in C_c^∞ if there is a compact K containing the supports of all ϕ_n and $\|\phi_n - \phi\|_{C^j(K)} \to 0$ for every j. Likewise a family $\phi(t) \in C_c^\infty$ converges to ϕ as $t \to t_0$ if there are a compact K and $\delta > 0$ with $\operatorname{supp}\phi(t) \subseteq K$ for $|t - t_0| < \delta$, and $\|\phi(t) - \phi\|_{C^j(K)} \to 0$ for every j.

This is a very strong notion. In particular, if $\phi_n \to \phi$ in C_c^∞ then $\langle T, \phi_n\rangle \to \langle T, \phi\rangle$ for every distribution T (subtract and apply Theorem 1.3.3). Much more is true.

Theorem 8.2.1 (Joint continuity)
If T is continuous in time and $\phi(t) \to \phi(t_0)$ in C_c^∞, then

$$\langle T(t), \phi(t)\rangle \xrightarrow{t\to t_0} \langle T(t_0), \phi(t_0)\rangle .$$

The same holds for sequences $T_n \to T$ in distribution and $\phi_n \to \phi$ in C_c^∞.

Proof. Write $\langle T(t), \phi(t)\rangle - \langle T(t_0), \phi(t_0)\rangle = \langle T(t), \phi(t) - \phi(t_0)\rangle + \langle T(t) - T(t_0), \phi(t_0)\rangle$. The second term tends to 0 by continuity in time. For the first, apply Lemma 8.1.1 on a compact interval around t_0 and the compact K containing all the supports: it is bounded by $C\|\phi(t) - \phi(t_0)\|_{C^j(K)} \to 0$. □

For a time-dependent test function $\phi : I \to C_c^\infty$, $\partial_t\phi(t)$ is the limit in C_c^∞ of $\big(\phi(t + h) - \phi(t)\big)/h$ as $h \to 0$, when it exists.

8.3 Calculus rules

Let S, T be time-dependent distributions, differentiable in time; "spacetime" means $\mathbb{R}^N \times I$.

- (Linearity) $\partial_t(aT + bS) = a\,\partial_t T + b\,\partial_t S$.
- (Leibniz rule for the pairing) If ϕ is a time-dependent test function that is differentiable in time, then
$$\frac{\mathrm{d}}{\mathrm{d}t}\langle T,\, \phi\rangle = \langle \partial_t T,\, \phi\rangle + \langle T,\, \partial_t \phi\rangle\,.$$
Proof: split the difference quotient as $\left\langle \frac{T(t+h)-T(t)}{h},\, \phi(t)\right\rangle + \left\langle T(t+h),\, \frac{\phi(t+h)-\phi(t)}{h}\right\rangle$ and apply Theorem 8.2.1 to the second term.
- (Product rule) If f is smooth in spacetime, then $\partial_t(fT) = (\partial_t f)T + f\,\partial_t T$. (Apply the Leibniz rule with test function $f(t)\phi$.)
- (Convolution I) If f is smooth in spacetime and compactly supported in space, then $\partial_t(T * f) = (\partial_t T) * f + T * (\partial_t f)$.
- (Convolution II) If $S(t)$ is compactly supported for all t (uniformly on compact time intervals), then $\partial_t(T * S) = (\partial_t T) * S + T * (\partial_t S)$.
- (Fourier) For tempered T, $\widehat{\partial_t T} = \partial_t \hat{T}$; this follows from $\left\langle \widehat{\partial_t T},\, \phi\right\rangle = \langle \partial_t T,\, \hat{\phi}\rangle = \frac{\mathrm{d}}{\mathrm{d}t}\langle T,\, \hat{\phi}\rangle$.
- If $\partial_t T = 0$ for all t, then T is constant in time (apply the one-variable mean value theorem to $t \mapsto \langle T(t),\, \phi\rangle$).

II

Applications

9 Ordinary Differential Equations and Liouville's Theorem

9.1 Linear ODEs via the Fourier transform

Consider the spring equation with $k = 9$,

$$u'' + 9u = 0.$$

A common approach is to take Fourier transforms. To do this rigorously we *assume* that u is tempered (does not grow too fast). Then

$$(2\pi i\xi)^2\hat{u} + 9\hat{u} = 0, \qquad \text{i.e.} \qquad (9 - 4\pi^2\xi^2)\,\hat{u} = 0.$$

The factor $(3 - 2\pi\xi)(3 + 2\pi\xi)$ has simple zeros at $\xi = \pm\frac{3}{2\pi}$ and no others, so by Theorem 4.3.5 $\hat{u} = A\delta_{-3/(2\pi)} + B\delta_{3/(2\pi)}$. Inverting,

$$u = Ae^{-3ix} + Be^{3ix}.$$

Real solutions are obtained by choosing $A, B \in \mathbb{C}$ appropriately, which gives the familiar combinations of $\cos 3x$ and $\sin 3x$.

⚠ We have only shown that these are the unique *tempered* solutions. The method is blind to solutions that grow faster than any polynomial.

This blindness can be severe. For $u' - u = 0$, the transform gives $(2\pi i\xi - 1)\hat{u} = 0$. For real ξ the factor never vanishes, so $\hat{u} = 0$ and $u = 0$: the *only tempered* solution is trivial, and the obvious solution e^x is invisible.

So the Fourier method should not be trusted to find *all* solutions. It is excellent, however, for finding *particular* solutions. Suppose we want to solve

$$Lu = f, \qquad L = \sum_{j=0}^{k} c_j D^j, \quad Du := u'.$$

Instead of attacking this directly, first solve

$$L\Phi = \delta_0.$$

Such a Φ is a ***fundamental solution***. If we have one, then $u = f * \Phi$ solves the original equation (whenever the convolution makes sense), because

$$Lu = L(f * \Phi) = f * (L\Phi) = f * \delta_0 = f.$$

To find Φ, take Fourier transforms: $P(\xi)\hat{\Phi} = 1$, where $P(\xi) = \sum_j c_j(2\pi i\xi)^j$ is the ***symbol*** of L. We would like to divide by P; this is delicate if P has real zeros, but with care one finds $\hat{\Phi} = 1/P(\xi)$ (suitably interpreted, e.g. as a principal value) plus "lower-order terms" supported on the zeros, and then inverts.

Example 9.1.1
Solve $u'' - u = \sin x$. First find Φ with $\Phi'' - \Phi = \delta_0$. The transform gives $(-4\pi^2\xi^2 - 1)\hat{\Phi} = 1$, and since the symbol has no real zeros,

$$\hat{\Phi} = \frac{-1}{1 + 4\pi^2\xi^2}.$$

Since $\widehat{e^{-|x|}} = \frac{2}{1+4\pi^2\xi^2}$, we get $\Phi = -\frac{1}{2}e^{-|x|}$. Therefore

$$u(x) = (\Phi * \sin)(x) = -\frac{1}{2}\int_{\mathbb{R}} e^{-|y|}\sin(x-y)\,dy = -\frac{1}{2}\sin x,$$

as one checks by computing $\int e^{-|y|}e^{-iy}\,dy = 2/(1+1) = 1$ and taking imaginary parts. Indeed $(-\frac{1}{2}\sin x)'' - (-\frac{1}{2}\sin x) = \sin x$.

In summary, distribution theory helps with constant-coefficient ODEs in two ways:

- it finds *all* solutions of $Lu = 0$ when every root of the characteristic polynomial is purely imaginary (equivalently, every zero of the symbol is real), since then every solution is tempered;
- it finds a particular solution of $Lu = f$ under suitable growth conditions on L and f.

9.2 Liouville's theorem

Recall Liouville's theorem: a bounded entire function is constant. Distribution theory gives a strikingly short proof.

Theorem 9.2.1 (Liouville)
Every bounded entire function $f : \mathbb{C} \to \mathbb{C}$ is constant. More generally, an entire function of polynomial growth is a polynomial.

Proof. Identify $\mathbb{C}$ with $\mathbb{R}^2$. By the Cauchy–Riemann equations, $\partial_{\bar{z}} f = 0$, i.e.

$$\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y} = 0.$$

A bounded (or polynomially bounded) function is tempered, so we may take Fourier transforms:

$$2\pi i(\xi_1 + i\xi_2)\,\hat{f} = 0.$$

The factor $\xi_1 + i\xi_2$ vanishes only at the origin, so by Proposition 4.2.1 and Theorem 4.2.2, $\hat{f} = \sum_{|\alpha|\le n} c_\alpha \partial^\alpha \delta_0$. Inverting, f is a polynomial in x and y. A bounded polynomial is constant, and a polynomially bounded one has bounded degree. □

10 Partial Differential Equations

10.1 Distributional solutions

There are many applications of Fourier analysis to classical PDEs; here we emphasize those that genuinely need distributions.

The ***transport equation*** $\partial_t u + a \cdot \nabla u = 0$ ($a \in \mathbb{R}^N$ fixed) is solved by the method of characteristics: $u(x,t) = f(x - at)$ for differentiable f. One often hears that u is also a solution when f is *not* differentiable, e.g. $u(x,t) = \mathbf{1}_{[-2,3]}(x - at)$. Morally this is right — the solution just transports its initial profile with velocity a — but literally $\partial_t u$ and ∇u do not exist. Such u are called ***distributional solutions***, and we can now say precisely what this means: u solves the PDE as a distribution on $\mathbb{R}^N \times \mathbb{R}$, where derivatives always exist.

Proposition 10.1.1
For every $f \in L^1_{\mathrm{loc}}(\mathbb{R}^N)$, $u(x,t) := f(x - at)$ solves $\partial_t u + a \cdot \nabla u = 0$ in the sense of distributions on $\mathbb{R}^N \times \mathbb{R}$.

Proof. For $\phi \in C_c^\infty(\mathbb{R}^N \times \mathbb{R})$,

$$\langle \partial_t u + a \cdot \nabla u,\, \phi \rangle = -\iint f(x - at)\,(\partial_t \phi + a \cdot \nabla \phi)(x,t)\,\mathrm{d}x\,\mathrm{d}t.$$

Substitute $x = y + at$ and put $\psi(y,t) := \phi(y + at, t)$, so that $\partial_t \psi(y,t) = (\partial_t \phi + a \cdot \nabla \phi)(y + at, t)$. The integral becomes $-\int f(y)\big(\int_{\mathbb{R}} \partial_t \psi(y,t)\,\mathrm{d}t\big)\,\mathrm{d}y = 0$, since $\psi(y,\cdot)$ has compact support. □

> *Remark 10.1.2. Weak solutions* are a closely related notion: one tests the equation against smooth functions (often including boundary or initial terms) and asks the resulting integral identity to hold. For linear equations like this one, being a distributional solution amounts to the same integral identity with compactly supported test functions.

One could also derive the solution by Fourier transform, either in spacetime or in space alone. Transforming in spacetime gets messy quickly. It is more natural to transform in space only, especially since time need not range over all of $\mathbb{R}$, and the cleanest way to organize this is through fundamental solutions.

10.2 Fundamental solutions

Two different objects go by the name "fundamental solution" in PDE:

- the Laplace kernel $C_N |x|^{2-N}$ (for $N \geq 3$), which satisfies $-\Delta\big(C_N |x|^{2-N}\big) = \delta_0$;
- the heat kernel $\Phi(x,t) = (4\pi t)^{-N/2} e^{-|x|^2/(4t)}$, which satisfies $\partial_t \Phi - \Delta \Phi = 0$ with $\Phi|_{t=0} = \delta_0$.

The delta appears in very different places because these two notions pursue different goals: the first describes the effect of a *forcing term*, the second the effect of an *initial condition*. We treat initial conditions first.

A typical linear problem has the form

$$\begin{cases} L_t u + L_x u = f, \\ u|_{t=0} = u_0, \quad \partial_t u|_{t=0} = u_1, \quad \dots, \quad \partial_t^{l-1} u|_{t=0} = u_{l-1}, \end{cases}$$

where:

- $u : \mathbb{R}^N \times I \to \mathbb{R}$, with I the time interval;
- $L_t = \sum_{k=0}^{l} c_k \partial_t^k$ is a constant-coefficient time operator of order l, typically ∂_t (heat) or ∂_t^2 (wave), and sometimes absent altogether (Laplace);
- L_x (often just L) is a spatial differential operator, which we *assume has constant coefficients*: $Lu = -\Delta u$ is allowed, $Lu = |x|^2 u$ is not;
- f is the forcing term, possibly time-dependent;
- $u_0, \dots, u_{l-1}$ are the l initial conditions.

10.2.1 One time derivative

Consider

$$\partial_t u + Lu = 0, \qquad u|_{t=0} = u_0.$$

Fundamental solution for an initial condition

The fundamental solution is a time-dependent distribution Φ solving

$$\partial_t \Phi + L\Phi = 0, \qquad \Phi|_{t=0} = \delta_0$$

in the sense of distributions.

Once Φ is known, the problem is solved for every initial condition: $u := \Phi * u_0$ (whenever the convolution makes sense, e.g. u_0 compactly supported). Indeed, by the calculus of Chapter 8,

$$\begin{aligned} \partial_t u + Lu &= (\partial_t \Phi) * u_0 + (L\Phi) * u_0 = (\partial_t \Phi + L\Phi) * u_0 = 0, \\ u|_{t=0} &= \Phi|_{t=0} * u_0 = \delta_0 * u_0 = u_0. \end{aligned}$$

The initial condition works precisely because δ_0 is the identity for convolution. (Convince yourself that ∂_t and L each pass onto Φ — for slightly different reasons.)

Example 10.2.1 (Transport)

The fundamental solution solves $\partial_t \Phi + a \cdot \nabla \Phi = 0$, $\Phi|_{t=0} = \delta_0$. Assume Φ is tempered and transform in space: $\partial_t \hat{\Phi} + 2\pi i\, a \cdot \xi\, \hat{\Phi} = 0$. Then $\partial_t \left(e^{2\pi i a \cdot \xi t} \hat{\Phi} \right) = 0$, so $e^{2\pi i a \cdot \xi t} \hat{\Phi}$ is constant in time and equals its value $\hat{\delta}_0 = 1$ at $t = 0$. Hence

$$\Phi = \left(e^{-2\pi i (at) \cdot \xi} \right)^{\vee} = \delta_{at}, \qquad \boxed{\Phi = \delta_{at}}$$

This is the only tempered fundamental solution. It says exactly what transport should do: a unit mass at the origin moves with velocity a. For initial data u_0,

$$u = \Phi * u_0 = \delta_{at} * u_0 = u_0(x - at),$$

the classical solution.

Example 10.2.2 (Heat)
The fundamental solution solves $\partial_t \Phi - \Delta \Phi = 0$, $\Phi|_{t=0} = \delta_0$. Here the tempered ansatz matters: pathological non-tempered solutions of the heat equation exist, and the ansatz excludes them. Transforming in space, $\partial_t \hat{\Phi} + 4\pi^2 |\xi|^2 \hat{\Phi} = 0$ with $\hat{\Phi}|_{t=0} = 1$, so

$$\hat{\Phi} = e^{-4\pi^2 |\xi|^2 t}.$$

To invert, recall that $\left(e^{-\pi s|x|^2}\right)^\wedge = s^{-N/2} e^{-\pi|\xi|^2/s}$ for $s > 0$. Writing $e^{-4\pi^2|\xi|^2 t} = e^{-\pi(4\pi t)|\xi|^2}$ and taking $s = 4\pi t$,

$$\boxed{\Phi(x,t) = \frac{1}{(4\pi t)^{N/2}} e^{-|x|^2/(4t)}}$$

This describes how a unit mass of heat spreads out, and for initial data u_0,

$$u(x,t) = (\Phi * u_0)(x,t) = \int_{\mathbb{R}^N} u_0(y) \frac{1}{(4\pi t)^{N/2}} e^{-|x-y|^2/(4t)} \, dy,$$

the familiar formula.

10.2.2 Several time derivatives

With l time derivatives there are l initial conditions, and correspondingly l fundamental solutions, one to tame each initial condition.

For $L_t u + L_x u = 0$ with L_t of order l, the fundamental solutions $\Phi_0, \dots, \Phi_{l-1}$ are defined by

$$L_t \Phi_k + L_x \Phi_k = 0, \qquad \partial_t^j \Phi_k|_{t=0} = \begin{cases} \delta_0, & j = k, \\ 0, & j \neq k, \end{cases} \qquad 0 \le j \le l-1.$$

Proposition 10.2.3
The solution of $L_t u + L_x u = 0$ with $\partial_t^j u|_{t=0} = 0$ for $j \neq k$ and $\partial_t^k u|_{t=0} = u_k$ is $u = u_k * \Phi_k$. Consequently the solution with general initial data $\partial_t^j u|_{t=0} = u_j$, $0 \le j \le l-1$, is

$$u = \sum_{j=0}^{l-1} u_j * \Phi_j.$$

Proof. All derivatives fall on Φ_k: $(L_t + L_x)(u_k * \Phi_k) = u_k * (L_t + L_x)\Phi_k = 0$ and $\partial_t^j (u_k * \Phi_k)|_{t=0} = u_k * \partial_t^j \Phi_k|_{t=0}$, which is u_k for $j = k$ and 0 otherwise. The second statement follows by linearity. □

Example 10.2.4 (The wave equation in one dimension)
Consider $\partial_t^2 u - c^2 \partial_x^2 u = 0$, $u|_{t=0} = u_0$, $\partial_t u|_{t=0} = u_1$, with wave speed $c > 0$ (mathematically removable, but kept for insight). In higher dimensions the computation is much harder. There are two fundamental solutions, with initial data $(\delta_0, 0)$ and $(0, \delta_0)$

respectively.

Φ_0. Assuming temperedness and transforming in x, $\partial_t^2\hat{\Phi}_0 + 4\pi^2c^2\xi^2\hat{\Phi}_0 = 0$. Imitating the solution of $f'' + k^2 f = 0$, we write

$$\hat{\Phi}_0 = A(\xi)e^{-2\pi ic\xi t} + B(\xi)e^{2\pi ic\xi t} \qquad (*)$$

with A, B distributions in ξ that do *not* depend on t. The initial data give $A + B = 1$ and $2\pi ic\xi(B - A) = 0$. The second equation says $\xi(B - A) = 0$, so $B - A = a\delta_0$ for some constant a, and $A = \frac{1-a\delta_0}{2}$, $B = \frac{1+a\delta_0}{2}$. Since $e^{\pm 2\pi ic\xi t}\delta_0 = \delta_0$, the δ_0 terms in $(*)$ cancel, leaving $\hat{\Phi}_0 = \frac{1}{2}e^{-2\pi ic\xi t} + \frac{1}{2}e^{2\pi ic\xi t}$. Inverting,

$$\boxed{\Phi_0 = \tfrac{1}{2}\delta_{-ct} + \tfrac{1}{2}\delta_{ct}}$$

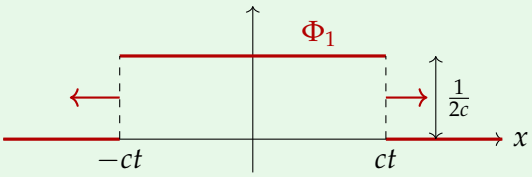
An initial displacement at rest splits into two equal halves travelling with velocities $\pm c$. This describes the effect of perturbing u_0.

Φ_1. The same computation gives $\hat{\Phi}_1 = Ae^{-2\pi ic\xi t} + Be^{2\pi ic\xi t}$, now with $A + B = 0$ and $2\pi ic\xi(B - A) = 1$. Thus $A = -B$ and $4\pi ic\xi B = 1$, i.e. $2\pi i\xi B = \frac{1}{2c}$ and $2\pi i\xi A = -\frac{1}{2c}$. Rather than divide by ξ, invert first, $\Phi_1 = \check{A} * \delta_{ct} + \check{B} * \delta_{-ct}$, and differentiate in x:

$$\partial_x\Phi_1 = (2\pi i\xi A)^\vee * \delta_{ct} + (2\pi i\xi B)^\vee * \delta_{-ct} = -\tfrac{1}{2c}\delta_{ct} + \tfrac{1}{2c}\delta_{-ct} = \frac{\delta_{-ct} - \delta_{ct}}{2c}.$$

A distribution whose derivative is a combination of deltas is a step function (Example 3.7.3 and Proposition 3.7.6): Φ_1 jumps up by $\frac{1}{2c}$ at $x = -ct$, down by $\frac{1}{2c}$ at $x = ct$, and is otherwise constant. Since $\Phi_1|_{t=0} = 0$, the constant is 0, so

$$\boxed{\Phi_1 = \frac{1}{2c}\,\mathbf{1}_{[-ct,ct]}}$$



A "poke" of unit impulse at the origin raises the string by $\frac{1}{2c}$ nearby, and this effect spreads at speed c in both directions.

Combining,

$$\begin{aligned} u = u_0 * \Phi_0 + u_1 * \Phi_1 &= u_0 * \frac{\delta_{ct} + \delta_{-ct}}{2} + u_1 * \frac{\mathbf{1}_{[-ct,ct]}}{2c} \\ &= \frac{u_0(x - ct) + u_0(x + ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} u_1(y)\,\mathrm{d}y, \end{aligned}$$

which is exactly ***d'Alembert's formula***.

10.3 Fundamental solutions for forcing terms

Forcing terms matter both for evolution equations ($\partial_t u - \Delta u = f$) and for stationary ones ($-\Delta u = f$). Since initial conditions play no role, the asymmetry between space and time disappears and we may simply consider $Lu = f$ on "spacetime", now regarded as

space. (If time is restricted to an interval, one can no longer take the Fourier transform in it.)

Forcing fundamental solution
A ***fundamental solution*** of $Lu = f$ is a distribution Φ with $L\Phi = \delta_0$.

It captures the response to a unit point force. If Φ is found, then $u = f * \Phi$ solves $Lu = f$, since $L(f * \Phi) = f * L\Phi = f * \delta_0 = f$. Fundamental solutions are not unique: if $-\Delta\Phi = \delta_0$, then also $-\Delta(\Phi + 1) = \delta_0$. This reflects the non-uniqueness of solutions of $Lu = f$ in the absence of boundary conditions.

Example 10.3.1 (The one-dimensional Laplacian)
Solve $-\Phi'' = \delta_0$ without Fourier analysis. Since $H' = \delta_0$ for the Heaviside function $H = \mathbf{1}_{(0,\infty)}$, Proposition 3.7.6 gives $-\Phi' = H + c_1$, which is a function, so integrating,

$$-\Phi(x) = x\mathbf{1}_{x>0} + c_1 x + c_2.$$

These are all fundamental solutions. The choice $c_1 = -\frac{1}{2}$, $c_2 = 0$ gives the symmetric one, $\Phi(x) = -\frac{1}{2}|x|$.

This Φ is not integrable, so, for instance, $u = 1 * \left(-\frac{1}{2}|x|\right)$ is meaningless as a solution of $-u'' = 1$. It might seem that such fundamental solutions are of little use. They are, in fact, extremely useful, as the next section shows.

10.4 Elliptic regularity

Distribution theory shows that solutions of $-\Delta u = 0$ are automatically smooth, and the key is the *existence* of a fundamental solution. We work on a domain Ω rather than on $\mathbb{R}^N$.

Theorem 10.4.1 (Fundamental solutions give smoothness)
Let L have constant coefficients and suppose there is a fundamental solution Φ, $L\Phi = \delta_0$, which is a smooth function on $\mathbb{R}^N \setminus \{0\}$. If $\Omega \subseteq \mathbb{R}^N$ is open, $f \in C^\infty(\Omega)$, and $u \in \mathcal{D}'(\Omega)$ solves $Lu = f$ in Ω, then u is a smooth function on Ω.

In the language of the subject: the hypothesis says that the *singular support* of Φ is $\{0\}$, and the conclusion, "Lu smooth implies u smooth", says that L is ***hypoelliptic***. The prototype is the Laplacian: $C_N|x|^{2-N}$ is not smooth, but it is away from the origin.

Proof. Smoothness is local, so fix $x_0 \in \Omega$ and $r > 0$ with $B(x_0, 2r) \subseteq \Omega$. Pick $\chi \in C_c^\infty(B(x_0, 2r))$ with $\chi \equiv 1$ on $B(x_0, r)$. Then χu is a compactly supported distribution on $\mathbb{R}^N$, and by the product rule

$$L(\chi u) = \chi f + w,$$

where w collects the terms in which at least one derivative falls on χ; hence w is supported in the annulus $A = \{r \le |x - x_0| \le 2r\}$. Since χu has compact support,

$$\chi u = \delta_0 * (\chi u) = (L\Phi) * (\chi u) = \Phi * L(\chi u) = \Phi * (\chi f) + \Phi * w.$$

The first term is smooth, as the convolution of a distribution with a C_c^∞ function. For the second, pick $\psi \in C_c^\infty(B(0, r/2))$ with $\psi \equiv 1$ near 0 and split $\Phi = \psi\Phi + (1 - \psi)\Phi$. The function $(1 - \psi)\Phi$ is smooth, so $\big((1 - \psi)\Phi\big) * w$ is smooth. And $(\psi\Phi) * w$ is supported in $B(0, r/2) + A$, which does not meet $B(x_0, r/2)$. Therefore on $B(x_0, r/2)$, $u = \chi u$ agrees with a smooth function. □

Corollary 10.4.2
Solutions of Poisson's equation $-\Delta u = f$ with f smooth are smooth. In particular every harmonic distribution is a smooth harmonic function.

(Take for granted — or verify — that $C_N|x|^{2-N}$ is a fundamental solution of $-\Delta$ for suitable C_N when $N \geq 3$; use $-\frac{1}{2\pi}\log|x|$ when $N = 2$.)

So regularity has been reduced to producing a fundamental solution that is smooth away from the origin. When do such fundamental solutions exist? A clean sufficient condition is ellipticity.

Symbol; ellipticity
The ***symbol*** of a constant-coefficient operator L is the polynomial P with $\widehat{Lu} = P(\xi)\hat{u}$. If P has degree m and P_m is its top-degree part (the ***principal symbol***), L is ***elliptic*** if $P_m(\xi) \neq 0$ for all real $\xi \neq 0$. Equivalently, there are $c, R > 0$ with $|P(\xi)| \geq c|\xi|^m$ for $|\xi| \geq R$.

Theorem 10.4.3
If L is elliptic, it has a fundamental solution Φ which is smooth on $\mathbb{R}^N \setminus \{0\}$.

Theorem 10.4.4 (Hypoellipticity of elliptic operators)
If L is elliptic and Lu is smooth, then u is smooth.

We do not have the tools to prove these here; Duistermaat–Kolk [2] is an excellent reference. The two key ideas are the *singular support* (the set of points near which a distribution fails to be smooth) and the *parametrix*, a useful weakening of a fundamental solution. As the definition of the symbol suggests, Fourier analysis is at the heart of it.

Example 10.4.5 (The bilaplacian)
$\Delta^2 u := \Delta(\Delta u) = \sum_{i,j} \partial_i^2 \partial_j^2 u$, so $\widehat{\Delta^2 u} = \sum_{i,j} 16\pi^4 \xi_i^2 \xi_j^2 \,\hat{u} = 16\pi^4|\xi|^4 \hat{u}$. The symbol $16\pi^4|\xi|^4$ is homogeneous of degree 4 and vanishes only at $\xi = 0$, so Δ^2 is elliptic, and every solution of $\Delta^2 u = f$ with f smooth is smooth. In particular biharmonic functions are smooth — which is not surprising: if $\Delta^2 u = 0$ then Δu is harmonic, hence smooth, and then u is smooth by the Laplacian case.

We close with an application in the spirit of Liouville's theorem.

Theorem 10.4.6
If $-\Delta u = 0$ on $\mathbb{R}^N$ and $|u(x)| \leq C(1 + |x|^n)$, then u is a polynomial of degree at most n.

Proof. u is tempered, and $4\pi^2|\xi|^2\hat{u} = 0$. By Proposition 4.2.1, $\operatorname{supp}\hat{u} \subseteq \{0\}$, so $\hat{u}$ is a finite combination of derivatives of δ_0 and u is a polynomial. The growth bound forces its degree to be at most n. □

11 Further Applications

11.1 The Poisson summation formula

Theorem 11.1.1 (Poisson summation)
For every $f \in \mathcal{S}(\mathbb{R})$,

$$\sum_{n\in\mathbb{Z}} \hat{f}(n) = \sum_{n\in\mathbb{Z}} f(n).$$

Proof. Let $\mathcal{C} := \sum_{n\in\mathbb{Z}} \delta_n$ be the ***Dirac comb***, a tempered distribution. It suffices to show $\hat{\mathcal{C}} = \mathcal{C}$, because then $\sum_n f(n) = \langle \mathcal{C}, f\rangle = \langle \hat{\mathcal{C}}, f\rangle = \left\langle \mathcal{C}, \hat{f}\right\rangle = \sum_n \hat{f}(n)$.

The comb is 1-periodic: $\tau_1\mathcal{C} = \mathcal{C}$. Taking Fourier transforms, $e^{-2\pi i\xi}\hat{\mathcal{C}} = \hat{\mathcal{C}}$, i.e.

$$\left(e^{-2\pi i\xi} - 1\right)\hat{\mathcal{C}} = 0.$$

The zeros of $e^{-2\pi i\xi} - 1$ are the integers, all simple, so Theorem 4.3.5 gives $\hat{\mathcal{C}} = \sum_n c_n\delta_n$.

It remains to show every $c_n = 1$. First, $\hat{\mathcal{C}}$ is periodic: since $\operatorname{supp}\mathcal{C} \subseteq \mathbb{Z}$ and $e^{2\pi in} = 1$, we have $(e^{2\pi ix} - 1)\mathcal{C} = 0$, and multiplication by $e^{2\pi ix}$ becomes translation by 1 on the Fourier side, so $\tau_1\hat{\mathcal{C}} = \hat{\mathcal{C}}$. Hence all c_n equal a common value c. Testing against the self-dual Gaussian $g(x) = e^{-\pi x^2}$,

$$c\sum_{n\in\mathbb{Z}} e^{-\pi n^2} = \langle \hat{\mathcal{C}}, g\rangle = \langle \mathcal{C}, \hat{g}\rangle = \sum_{n\in\mathbb{Z}} e^{-\pi n^2},$$

so $c = 1$. □

The same argument proves the N-dimensional version $\sum_{n\in\mathbb{Z}^N} f(n) = \sum_{n\in\mathbb{Z}^N} \hat{f}(n)$ for $f \in \mathcal{S}(\mathbb{R}^N)$.

11.2 Probability

The ***characteristic function*** of a random variable X is $\varphi_X(t) := \mathbb{E}e^{itX} = \int_{\mathbb{R}} e^{itx}\,\mathrm{d}\mu_X(x)$, which is essentially the Fourier transform of the law: $\varphi_X(t) = \check{\mu}_X(t/2\pi)$. Laws are probability measures, hence finite, so $\mu_X * \mu_Y$ is a probability measure (Exercise E7). When X and Y are independent, $\mu_{X+Y} = \mu_X * \mu_Y$, and taking Fourier transforms gives $\hat{\mu}_{X+Y} = \hat{\mu}_X\hat{\mu}_Y$, i.e. the classical identity $\varphi_{X+Y} = \varphi_X\varphi_Y$.

Distribution theory may not be a workhorse in probability, but it unifies characteristic functions with the general Fourier theory rather nicely.

11.3 The Weierstrass approximation theorem

Theorem 11.3.1 (Weierstrass)
Let $K \subseteq \mathbb{R}^N$ be compact. Every continuous $f : K \to \mathbb{R}$ is a uniform limit of polynomials on K.

Proof. Extend f to a continuous, compactly supported F on $\mathbb{R}^N$ (Tietze), with $\operatorname{supp} F \subseteq B(0,R)$ and $K \subseteq B(0,R)$. Let $G_\varepsilon(x) := \varepsilon^{-N} e^{-\pi|x|^2/\varepsilon^2}$; this is an approximation to the identity ($\int G_\varepsilon = 1$, concentrating at 0), and since F is uniformly continuous, $F * G_\varepsilon \to F$ uniformly. Fix ε with $\|F * G_\varepsilon - F\|_\infty < \eta/2$.

For $x \in K$ and $y \in \operatorname{supp} F$ we have $|x-y| \le 2R$. The power series of e^{-s} converges uniformly for $0 \le s \le \pi(2R/\varepsilon)^2$, so there is a polynomial Q with $|e^{-\pi|z|^2/\varepsilon^2} - Q(|z|^2)| < \eta/(2\varepsilon^{-N}\|F\|_{L^1})$ for $|z| \le 2R$. Then

$$P(x) := \varepsilon^{-N} \int F(y)\, Q(|x-y|^2)\, \mathrm{d}y$$

is a polynomial in x and satisfies $|P - F * G_\varepsilon| < \eta/2$ on K. Hence $|P - f| < \eta$ on K. □

11.4 Perimeter

Finally, distributions shed light on the notion of perimeter in geometric measure theory. They are not needed to develop that theory (see Maggi [4]), but they explain where the definition comes from.

For a reasonable set $E \subseteq \mathbb{R}^N$, the perimeter is intuitively the size of its boundary, i.e. the amount of "variation" of the indicator $\mathbf{1}_E$. This suggests looking at the gradient $-\nabla \mathbf{1}_E$. Since $\mathbf{1}_E$ is not differentiable, the gradient is taken in the distributional sense: it has N components $-\partial_i \mathbf{1}_E$, with

$$\langle -\partial_i \mathbf{1}_E,\, \phi \rangle = \langle \mathbf{1}_E,\, \partial_i \phi \rangle = \int_E \partial_i \phi \, \mathrm{d}x.$$

Thus $-\nabla \mathbf{1}_E$ is a vector-valued distribution, acting on vector fields $\Phi \in C_c^\infty(\mathbb{R}^N; \mathbb{R}^N)$ by

$$\langle -\nabla \mathbf{1}_E,\, \Phi \rangle = \langle \mathbf{1}_E,\, \operatorname{div} \Phi \rangle = \int_E \operatorname{div} \Phi \, \mathrm{d}x.$$

For a smooth bounded set, the divergence theorem turns this into $\int_{\partial E} \Phi \cdot \nu \, \mathrm{d}\mathcal{H}^{N-1}$: the distribution $-\nabla \mathbf{1}_E$ is the measure $\nu\, \mathcal{H}^{N-1} \llcorner \partial E$, the *outward unit normal* spread over the boundary. This is the reason for the minus sign.

If $-\nabla \mathbf{1}_E$ is a (vector-valued, finite) measure — equivalently $\mathbf{1}_E \in BV$ — the perimeter can be measured, and is defined as its total variation. By duality, the total variation of a vector measure μ is

$$\|\mu\| = \sup \left\{ \int \Phi \cdot \, \mathrm{d}\mu : \Phi \in C_c^\infty(\mathbb{R}^N; \mathbb{R}^N),\ \|\Phi\|_\infty \le 1 \right\} = \sup_{\|\Phi\|_\infty \le 1} \langle \mu,\, \Phi \rangle .$$

Therefore

$$\begin{aligned} \operatorname{Per} E = \|-\nabla \mathbf{1}_E\| &= \sup_{\|\Phi\|_\infty \le 1} \langle \mathbf{1}_E,\, \operatorname{div} \Phi \rangle \\ &= \sup \left\{ \int_E \operatorname{div} \Phi \, \mathrm{d}x : \Phi \in C_c^\infty(\mathbb{R}^N; \mathbb{R}^N),\ \|\Phi\|_\infty \le 1 \right\}, \end{aligned}$$

which is precisely the definition of perimeter used in geometric measure theory.

12 Exercises and Problems

12.1 Exercises

These are routine checks of understanding.

E1. Show that the adjoint of the Fourier transform (on $\mathcal{S}$, with respect to the bilinear pairing $\int uv$) is the Fourier transform itself, and with respect to the L^2 inner product it is the inverse transform.

E2. Show that convolution of finite measures is commutative.

E3. For tempered T, find a sensible definition of the conjugate $\overline{T}$ and prove that $\widehat{\overline{T}} = \overline{\check{T}}$.

E4. Using Fourier methods, find all solutions of $u'''(x) = 1$ with at most polynomial growth. Does the method rule out exponential solutions?

E5. Verify from the definition of convolution that $T * \delta_0 = T$.

E6. Verify $T * \delta_0 = T$ for tempered T using the Fourier transform.

E7. Show that the convolution of two probability measures is a probability measure.

E8. Without using fundamental solutions, prove that every tempered distribution u solving the Helmholtz equation $\Delta u + 4\pi^2 u = 0$ is a smooth function. *Hint: where is $\hat{u}$ supported?*

E9. Prove that every distribution u solving the Helmholtz equation is smooth.

E10. Complete Exercise 4.1.2: solve $xT = S$ for a given distribution S.

12.2 Problems

These require a little more thought, and showcase some facts that did not fit elsewhere.

P1. Using only the definitions up to differentiation, and the fact that the solutions of $xT = 0$ are $T = a\delta_0$, find all solutions of $x^2 T = 0$ (without Theorem 4.2.2).

P2. Show that $\hat{\hat{\hat{f}}} = f$ for all $f \in \mathcal{S}(\mathbb{R}^N)$.

P3. Let $\mu = \sum_{n\in\mathbb{Z}} \delta_n$. Show that $\mu' = \sum_{n\in\mathbb{Z}} \delta_n'$ and compute $\mu' * \mathbf{1}_{[0,1]}$.

P4. Show that $f_n \to f$ almost everywhere does not imply $f_n \to f$ in distribution.

P5. Find the unique tempered solution of $f' - f = \delta_0$, and deduce a particular solution of $f' - f = g$ for $g \in C(\mathbb{R})$ (with suitable growth).

P6. If T is compactly supported, show that $\langle T, \phi\rangle$ makes sense for every $\phi \in C^\infty$ (not necessarily compactly supported). Deduce that the definition $(S * T) * \phi = S * (T * \phi)$ also works when S, rather than T, is compactly supported: then $T * \phi$ is merely C^∞, but S can be tested against it.

P7. Find a compactly supported $f \in L^1$ and a compactly supported distribution T such that $f * T$ is a distribution but not a measure.

P8. The heuristic "$f * g$ inherits the best regularity of f and g" is not entirely truthful for distributions. Find a compactly supported $f \in C_c^{100}(\mathbb{R})$ and a compactly supported distribution g such that $f * g$ is a discontinuous function.

P9. Find the error in the following "proof" that every harmonic function vanishes. Let $-\Delta u = 0$ and let Φ satisfy $-\Delta\Phi = \delta_0$. Then

$$u = \delta_0 * u = (-\Delta\Phi) * u = -\Delta(\Phi * u) = \Phi * (-\Delta u) = \Phi * 0 = 0.$$

P10. Let T be a distribution and $h_n \to 0$. Define $\langle T_n, \phi\rangle := \langle T, \phi(\cdot - h_n)\rangle$. Show that $T_n \to T$ in distribution.

P11. Suppose L has a fundamental solution Φ with *bounded singular support*: Φ is smooth on $\mathbb{R}^N \setminus B(0,R)$ for some R. Prove that for every smooth f on $\mathbb{R}^N$, every solution of $Lu = f$ on $\mathbb{R}^N$ is smooth.

12.3 Hints

- **P1.** If $x^2T = 0$, then $xT = a\delta_0$ for some a. Find one particular solution S of $xS = a\delta_0$ (try a multiple of δ_0'), and then $x(T - S) = 0$.
- **P9.** Which of the convolutions are actually defined, and which associativity law is being used? Neither Φ nor u need be compactly supported.
- **E4.** On the Fourier side the equation reads $(2\pi i\xi)^3\hat{u} = \delta_0$ up to a constant; use Proposition 4.3.2 for the homogeneous part.

13 Further Reading

Several important topics were omitted, either for simplicity or because they go beyond the scope of these notes:

- *homogeneous distributions*, which have many useful properties;
- *tensor products* of distributions and the closely related *Schwartz kernel theorem*;
- the *singular support* and the *wavefront set*, which describe precisely where and in which directions a distribution is singular; among other things they allow one to multiply distributions in more situations than those discussed here;
- *microlocal analysis* and *pseudodifferential operators*, which use Fourier analysis to generalize differentiation far beyond constant-coefficient operators;
- *hyperfunctions*, an even more general notion of generalized function;
- the general *division problem*: solving $fT = S$ when f has zeros of infinite order or vanishes on large sets (e.g. dividing by x^2 in general, or by $1 - |x|^2$), which the source series set aside for a future installment.

Recommended texts:

- Hörmander, *The Analysis of Linear Partial Differential Operators I* [1] — the standard reference;
- Duistermaat and Kolk, *Distributions: Theory and Applications* [2] — much gentler, and an excellent place to learn the theory;
- Rudin, *Functional Analysis* [3], Part II — a concise functional-analytic treatment.

Bibliography

[1] L. Hörmander, *The Analysis of Linear Partial Differential Operators I: Distribution Theory and Fourier Analysis*, 2nd ed., Springer, 1990.

[2] J. J. Duistermaat and J. A. C. Kolk, *Distributions: Theory and Applications*, Birkhäuser, 2010.

[3] W. Rudin, *Functional Analysis*, 2nd ed., McGraw–Hill, 1991.

[4] F. Maggi, *Sets of Finite Perimeter and Geometric Variational Problems*, Cambridge University Press, 2012.

[5] D. Yu, `dylan.sty` LaTeX style file, https://github.com/Yu-Dylan/tex/blob/master/dylan.sty.